# THE DIRECT-DECORATION ROOTED-END COMPACTIFICATION: AN EXACT ONE-SCALE MODEL FOR ROOTED COMPONENT PACKAGES

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Abstract. Let $f$ be a quadratic polynomial at a non-principal direct-decoration scale. Assume the relevant Kahn–Lyubich puzzle and dynasty data exist, the explicit finite-depth graph-avoidance input (2.5) holds, and the modified principal nest terminates in a primitive quadratic-like renormalization. The resulting separator $P$, iterate $g = f^q$, disjoint closed univalent pullbacks $Q_0, Q_1 \subset P$, and terminal primitive root $K_\epsilon$ satisfy an abstract four-hypothesis package. Under that package, the finite fibers $K_w = \phi_w(K_\epsilon)$ and nested end fibers $E_\omega = \bigcap_m Q_{\omega|m}$ form a compact planar set $\mathcal{H}_\partial \subset K(f)$. Its connected components are exactly these fibers, and its component quotient is the binary address compactification $\Omega_\partial$. Off the root, $g$ is a trivial two-sheeted covering of $\mathcal{H}_\partial$ semiconjugate to first-symbol deletion; geometric last-symbol deletion is a different parent map, and the root has its own terminal-cycle transition.

Every fiber is full and $\widehat{\mathbb{C}} \setminus \mathcal{H}_\partial$ is connected. Pairwise disjoint Jordan gap cells permit replacement over an arbitrary, possibly infinite, set of finite addresses by nonempty connected continua while compactness, the exact component structure, and the continuous quotient persist; full replacements preserve nonseparation. Finitely many protected root prime ends transport to a full finite carrier meeting $\mathcal{H}_\partial$ in exactly the selected fibers and preserving their cyclic order. Thus the compact packet layer admits infinite modification even though the protected routing layer remains finite. The package is inclusion-natural under root enlargement in the gap. A documented deeper pairwise-disjoint pullback construction gives the same abstract theory with transition time $nq$ if the exact $q$-level graph input is not invoked.

This is a positive control for virtual renormalization, not a construction of the virtual Cantor package: no identification of end fibers with small Julia sets, second satellite portrait, analytic self-return on an end fiber, or uniform incidence bound is claimed.

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## 1. Introduction

1.1. **The rooted-end problem.** The MLC conjecture—local connectivity of the Mandelbrot set—has been reduced, over three decades, to renormalization statements of increasingly combinatorial character [DH85, Hub93, Lyu97, Kah06, KL08, KL09, KKL26]. The present frontier is unbounded satellite combinatorics: traditional complex a priori bounds cannot be uniform along satellite relative periods tending to infinity [BLOT22], and the current proposals replace bounds by structural transition statements for towers of renormalizations. In Dudko's program [Dud25], the central objects are *virtual Julia sets*: connected hulls of Cantor families of small Julia sets carrying a finite part of the critical orbit. Kahn–Kapiamba–Lyubich [KKL26] construct, in their native primitive category, a positive control: a Hubbard tree inside the full dynamical carrier, lifted from the locally connected rational lamination model when the carrier itself is not locally connected.

Any such program must eventually manipulate a *rooted component package*: a distinguished root continuum, a family of homeomorphic copies of it indexed by finite dynamical addresses, limit fibers indexed by infinite addresses, transition dynamics between them, and access data (prime ends, protected collars, connectors) that survive the passage to limits. The sources used by the present program do not supply, in one theorem, a compact planar realization carrying all of the component, parent, root-arrow, and protected-access data listed here, and several plausible shortcuts are provably underdetermined (Section 8).

This paper constructs one such package completely, at one concrete scale, with every hypothesis displayed and every typing arrow realized by an actual restriction of an iterate of the polynomial. The setting is the *first direct-decoration scale* of Kahn–Lyubich [KL08, KL09]: a quadratic polynomial $f$ chosen in a non-principal decoration outside a directly attached satellite

copy of the Mandelbrot set, whose modified principal nest terminates in a primitive quadratic-like renormalization. There the puzzle supplies a separator piece $P$, an iterate $g = f^q$, and two disjoint closed univalent pullbacks $Q_0, Q_1 \subset P$ under the finite-depth graph-avoidance input (2.5); the nest supplies a terminal primitive root continuum $K_\epsilon$ lying in the first-level gap $P \setminus (Q_0 \cup Q_1)$.

1.2. **Main results.** Write $\phi_a \colon P \to Q_a$ $(a \in \{0,1\})$ for the inverse branches of $g$, $\phi_w = \phi_{a_1} \circ \cdots \circ \phi_{a_m}$ for a finite binary word $w = a_1 \cdots a_m$, and

$$Q_w = \phi_w(P), \qquad K_w = \phi_w(K_\epsilon), \qquad E_\omega = \bigcap_{m \geq 1} Q_{\omega|m} \quad (\omega \in \{0,1\}^{\mathbb{N}}).$$

The binary address compactification $\Omega_\partial$ consists of all finite words and all infinite words, topologized as a compact subset of $\{0,1,\#\}^{\mathbb{N}}$. The *rooted-end compactum* is

$$\mathcal{H}_\partial = \bigcup_w K_w \ \cup \ \bigcup_\omega E_\omega.$$

The main results, proved under the displayed Standing Hypotheses 2.2, are:

(I) *Compact component realization and quotient* (Theorems 5.3 and 5.4). $\mathcal{H}_\partial$ is a compact subset of $K(f)$ (of $J(f)$ under (H3)); its connected components are exactly the fibers $K_w$ and $E_\omega$; and the component projection $\pi_\partial \colon \mathcal{H}_\partial \to \Omega_\partial$ is a continuous closed surjection, so the component space of $\mathcal{H}_\partial$ is homeomorphic to $\Omega_\partial$. In particular the component decomposition is upper semicontinuous.

(II) *Exact two-branch partial shift* (Theorem 5.5). $\mathcal{H}_\partial \setminus K_\epsilon$ is the disjoint union of the clopen sheets $H_a = \mathcal{H}_\partial \cap Q_a = \phi_a(\mathcal{H}_\partial)$, and $g \colon \mathcal{H}_\partial \setminus K_\epsilon \to \mathcal{H}_\partial$ is a trivial two-sheeted covering semiconjugated by $\pi_\partial$ to the first-symbol deletion shift $\sigma_{\mathrm{pre}}$.

(III) *Two parent maps and a separate root arrow* (Remarks 5.6 and 5.7, Figure 1). Geometric nesting deletes the *last* symbol, $\mathrm{gpar}(wa) = w$; dynamical transport deletes the *first*, $\sigma_{\mathrm{pre}}(aw) = w$. These are different arrows with different domains, and the package keeps both. The root carries a third, separate transition $\tau_{\mathrm{root}}$.

(IV) *Fullness, nonseparation, and the end compactum* (Theorem 6.1 and Corollary 6.2). Every fiber is a full continuum, $\widehat{\mathbb{C}} \setminus \mathcal{H}_\partial$ is connected, and the end compactum $\mathcal{H}_\infty = \bigcup_\omega E_\omega$ is a compact nonseparating set whose components are the $E_\omega$, with component space the Cantor space $\{0,1\}^{\mathbb{N}}$ and with $g$ realizing the one-sided binary shift through two univalent branches.

(V) *Cellwise replacement at arbitrary addresses, and naturality* (Theorems 5.8, 6.4 and 6.5 and Corollary 6.7). The package admits an equivariant atlas of pairwise disjoint closed Jordan cells $D_w$; the fibers over an *arbitrary* set of finite addresses may be replaced by larger compact connected sets inside their cells while compactness, the exact component structure, and the continuous component quotient survive, and—when the replacements are full—nonseparation survives as well. The compactification package is natural under root enlargement within the gap, and a full enlarged root carries the entire fullness and cell theory.

(VI) *Finite protected carrier* (Theorem 7.2). Finitely many protected prime-end chains at the root transport through the inverse branches to finitely many selected fibers, and there is a full connected carrier $H$ with $H \cap \mathcal{H}_\partial = \bigcup_{w \in F} K_w$ exactly, in whose complement the transported chains survive as prime-end chains with their cyclic order.

The decisive separation is between two layers. The compact *packet layer* may contain infinitely many finite-address modifications, with no diameter control or coherence across addresses.

Only the protected *routing layer* remains finite, because its carrier is an embedded finite tree. This division is the main reusable advance beyond the one-scale compactification itself.

1.3. **What this paper is, and is not.** The abstract topology here is classical: a binary rooted tree with its ends, realized as nested disks with a Cantor limit set, is the oldest compactification in the subject. The contribution is not a new compactification theory. It is the *exact dynamical realization* of the typed package in one concrete complex-dynamical setting:

- the finite fibers are actual Julia-set pullbacks and the end fibers are actual nested puzzle intersections, in one compactum inside $K(f)$;
- both parent maps are retained and distinguished, and the dynamical one is realized by named univalent branches with time $q$, a recorded inverse sheet, and an exact two-sheeted covering off the root;
- the component space is computed exactly, as a continuous closed quotient;
- fullness, complement connectedness, the cell atlas, arbitrary-address replacement, and the protected carrier are proved for this realization, not for a symbolic model imposed afterwards;
- the compactification package is natural under root enlargement within the gap (Theorem 5.8 and Corollary 6.7), which is the enlargement the virtual program actually performs.

Equally important is what is *not* claimed; see Non-claims 2.3 and Section 8. This paper does not construct Dudko's virtual Cantor package. The end fibers are not identified with small Julia sets; no second satellite portrait, no analytic return map on an end fiber, and no uniform incidence bound is constructed. The honest role of the results is that of a positive control: a complete, directly visible, one-scale model showing what a correctly typed rooted-end component package looks like, against which proposed virtual constructions can be audited. The statements that remain open are recorded as Problems 8.2 and 8.3, and Appendix B records the adversarial boundary cases the proofs are required to pass.

1.4. **Organization.** Section 2 fixes the dynamical setting and states the standing hypotheses, together with the local pullback-boundary lemma that source-locks them. Section 3 collects the plane-topology toolkit (Čech–Alexander duality for disjoint full packets, nested-packet nonseparation, pointed Schoenflies transport, Jordan neighborhoods, slit and tree lemmas, and the crosscut-chain conventions). Section 4 constructs protected regulated connectors from certificates and proves the exact free-component criterion. Section 5 proves the compactification, quotient, and partial-shift theorems and the two-parent-map structure; Section 6 proves fullness, nonseparation, the end compactum, the gap-cell atlas, arbitrary-address replacement, and full-root naturality; Section 7 proves the finite protected-carrier theorem. Section 8 delimits scope and states the open problems. Appendix A records the concordance with the companion program manuscript, and Appendix B the adversarial boundary cases.

## 2. The first direct-decoration scale and the standing hypotheses

2.1. **Dynamical setting.** Let $f = f_c(z) = z^2 + c$ with $c$ in a non-principal decoration attached to a directly attached satellite copy $\mathcal{M}_{t/q}$, $q > 1$, of the Mandelbrot set, with $c$ outside the copy itself. In the Kahn–Lyubich decoration puzzle [KL08], let $Y^1$ denote the critical depth-one piece and $Z_j^1$ the off-critical depth-one pieces, and set

$$n = \min\{m \geq 1 : f^{mq}(0) \in \operatorname{int} Z_j^1 \text{ for some } j\}. \tag{2.1}$$

The minimum is finite because $c$ lies outside the satellite copy. The terminal component $Z^1_\kappa$ and the intermediate pullback components determine the escape route, and the selected escape piece is the critical piece

$$V_{\mathrm{esc}} = f^{-nq}\big(Z^1_\kappa\big)\big|_0 = Y^{nq+1}. \tag{2.2}$$

Kahn–Lyubich define this piece as the *first kingdom* and prove that every domain of its first-return map is compactly contained in it [KL08, KL09].

We assume that the puzzle and dynasty data used below are defined and that the modified principal nest built on (2.2) *terminates in a primitive quadratic-like renormalization*. This is the stopping condition of the Kahn–Lyubich dynasty construction [KL09]; the citation supplies provenance, while the paper assumes the resulting data rather than extending that source beyond its standing scope. The abstract results below require only Standing Hypotheses 2.2. The terminal datum is a polynomial-like first-return branch

$$f^p\colon A_0 \longrightarrow B, \qquad A_0 \Subset B, \tag{2.3}$$

of some period $p$, where $A_0$ is the critical first-return domain containing 0 [KL09, DH85]. Let $K_\epsilon$ denote its filled Julia set (the *terminal primitive root*) and let $\mathcal{K}_i = f^i(K_\epsilon)$, $0 \le i < p$, denote the terminal cycle, so $\mathcal{K}_0 = K_\epsilon$.

**Proposition 2.1** (Critical-base cycle exclusion)**.** *For $1 \le i < p$ we have $\mathcal{K}_i \cap B = \varnothing$. The complete cycle itinerary is $\mathcal{K}_1 \to \mathcal{K}_2 \to \cdots \to \mathcal{K}_{p-1} \to \mathcal{K}_0$.*

*Proof.* Equation (2.3) is a first-return branch, so $f^i(A_0) \cap B = \varnothing$ for $1 \le i < p$. Since $K_\epsilon \subset A_0$, the same holds for $\mathcal{K}_i = f^i(K_\epsilon)$. The itinerary follows from the definition of $\mathcal{K}_i$ and the complete invariance of the filled Julia set of the branch $f^p\colon A_0 \to B$. □

2.2. **The package data.** The compactification uses the following data extracted from the construction above. Let

$$P = Y^{(n-1)q+1}, \qquad g = f^q, \qquad g\colon Q_a \longrightarrow P \quad (a \in \mathtt{A} := \{0,1\}) \tag{2.4}$$

where $Q_0, Q_1$ are the two univalent pullbacks of $P$ under $g$ inside $P$ supplied by the decoration construction [KL08]: each $Q_a$ is the closure of a component of $g^{-1}(\operatorname{int} P)$, and the two components are distinct. Let $V^0 = Y^{nq+1}$ be the first kingdom (2.2), so that $K_\epsilon \Subset \operatorname{int} V^0$ and $V^0, Q_0, Q_1$ are distinct puzzle pieces of the common depth $nq+1$. Write $\phi_a\colon P \to Q_a$ for the inverse branches.

For the exact $q$-level realization we expose the only additional source-sensitive boundary input. Let $\Gamma^d$ be the depth-$d$ puzzle graph and assume

$$\{f^j(0) : j \ge 0\} \cap \Gamma^d = \varnothing \qquad (0 \le d \le nq+1). \tag{2.5}$$

Thus the critical orbit avoids every puzzle graph through the deepest level used here. We treat (2.5) as an explicit finite-depth source interface, not as a consequence hidden inside the word "puzzle." The source-certified fallback in Remark 2.6 removes this input at the cost of changing the transition time from $q$ to $nq$.

We isolate exactly the properties of these data that the proofs use.

**Standing Hypotheses 2.2.** Throughout the paper:

- (H1) $P$, $Q_0$, $Q_1$ are *closed* Jordan disks with $Q_0 \cup Q_1 \subset P$, and the two closed pieces are disjoint: $Q_0 \cap Q_1 = \varnothing$. Disjointness of the closed pieces, not merely of their interiors, is used.
- (H2) Each restriction $g|_{Q_a}$ is proper of degree one and extends to a homeomorphism $g\colon Q_a \to P$ of the closed Jordan pieces; equivalently, each $\phi_a\colon P \to Q_a$ is an orientation-preserving homeomorphism of closed Jordan disks.

(H3) $K(f) = J(f)$: the filled Julia set of $f$ has empty interior.
(H4) The root $K_\epsilon$ is a nondegenerate full continuum (a full continuum containing more than one point) with

$$K_\epsilon \subset \operatorname{int} P \setminus (Q_0 \cup Q_1); \tag{2.6}$$

it lies in the *first-level gap*. We write $G_0$ for the component of $\operatorname{int} P \setminus (Q_0 \cup Q_1)$ containing $K_\epsilon$.

**Non-claims 2.3.** Equally binding throughout:

(N1) The root transition $\tau_{\text{root}} \colon K_\epsilon \mapsto f^q(K_\epsilon)$ is separate data (Remark 5.7); it is *not* an extension of the first-symbol deletion map $\sigma_{\text{pre}}$, and no assertion of permanent escape from $\mathcal{H}_\partial$ is made.
(N2) No second satellite portrait is constructed, and the end fibers $E_\omega$ are *not* identified with small Julia sets.
(N3) No analytic return map *on* an end fiber is constructed: no first-return or polynomial-like self-return structure on any single $E_\omega$ is claimed. The analytic transitions $E_{a\omega} \to E_\omega$ of (5.5) *are* proved; they are a different assertion.
(N4) No uniform (period-independent) incidence or marking bound is claimed; see Proposition 8.1.

2.3. **Source-locking the hypotheses.** Closed-piece disjointness in (H1) is not formal vocabulary: closed puzzle pullbacks may in general touch at boundary points, as Kahn–Lyubich themselves caution [KL08], and any common point of $Q_0$ and $Q_1$ necessarily lies on both boundaries. The next lemma is the local statement that makes the exclusion argument exact.

**Lemma 2.4** (Local uniqueness of a regular closed pullback)**.** *Let $h$ be a holomorphic map defined near $z \in \mathbb{C}$ with $h'(z) \neq 0$, and let $P$ be a closed Jordan disk containing $h(z)$. Then $z$ belongs to the closure of at most one component of $h^{-1}(\operatorname{int} P)$.*

*Proof.* Choose an open topological disk $V$ about $h(z)$ and a neighborhood $W$ of $z$ such that $h \colon W \to V$ is a homeomorphism. If $h(z) \in \operatorname{int} P$, shrink $V$ so that $V \subset \operatorname{int} P$; then $V \cap \operatorname{int} P = V$ is connected. If $h(z) \in \partial P$, apply the Schoenflies theorem to a chart in which $\partial P$ near $h(z)$ is a straight segment and shrink $V$ to a round disk in that chart; then $V \cap \operatorname{int} P$ is an open half-disk, again connected. In both cases

$$W \cap h^{-1}(\operatorname{int} P) = (h|_W)^{-1}\big(V \cap \operatorname{int} P\big)$$

is connected. If a component $C$ of $h^{-1}(\operatorname{int} P)$ has $z \in \overline{C}$, then $C$ meets the connected set $W \cap h^{-1}(\operatorname{int} P)$, which is therefore contained in $C$. Two distinct components would both contain this nonempty connected set, contradicting disjointness of components. □

**Lemma 2.5** ($q$-level realization of the standing hypotheses)**.** *Under (2.5), the decoration data (2.4) satisfy (H1), (H2), and (H4).*

*Proof.* The pieces $P, Q_0, Q_1$ are closed Jordan puzzle pieces and the two pullbacks are closures of distinct components of $g^{-1}(\operatorname{int} P)$ [KL08]. Suppose the closed pieces $Q_0, Q_1$ met at a point $z$. Since the interiors of distinct depth-$(nq+1)$ pieces are disjoint complementary components of the depth-$(nq+1)$ puzzle graph, $z$ lies on that graph. We claim $z$ is a regular point of $g = f^q$. Indeed, the critical points of $g$ are the points $\zeta$ with $f^i(\zeta) = 0$ for some $0 \leq i < q$, and $f$ maps the depth-$d$ puzzle graph into the depth-$(d-1)$ graph; if $z$ were critical, $0 = f^i(z)$ would therefore lie on a puzzle graph. But (2.5) excludes that possibility. Hence $z$ is a regular point of $g$, and Lemma 2.4 puts $z$ in the closure of at most one component of $g^{-1}(\operatorname{int} P)$. Since $z \in Q_0 \cap Q_1$ and the $Q_a$ are closures of distinct components, this is a contradiction. This proves (H1).

A proper degree-one holomorphic map between Jordan domains is a conformal isomorphism of the interiors; since the pieces are closed Jordan disks bounded by finitely many ray and equipotential arcs, Carathéodory extension gives a homeomorphism of the closed pieces, orientation-preserving because the interior map is holomorphic. This proves (H2).

For (H4): the filled Julia set of the polynomial-like restriction (2.3) is a full continuum [DH85]. Nondegeneracy is automatic: $K_\epsilon$ is the filled Julia set of a degree-two polynomial-like branch, whose Julia set is a perfect set [DH85], so $K_\epsilon$ is not a point. The three pieces $V^0, Q_0, Q_1$ are distinct puzzle pieces of the common depth $nq+1$; the interior of each is a complementary component of the depth-$(nq+1)$ puzzle graph, so $\operatorname{int} V^0$ is disjoint from the closed pieces $Q_0$ and $Q_1$. Since $K_\epsilon \Subset \operatorname{int} V^0$ and $V^0 \subset P$ (deeper critical pieces are nested), we get (2.6). □

*Remark* 2.6 (Source audit and a documented fallback). The exact $q$-level proof of (H1) uses two distinct open $f^q$-pullback components from Kahn–Lyubich and the separately stated input (2.5). This is the only source-sensitive boundary gate in the paper. A source-certified fallback avoids it: Kahn–Lyubich construct, for every vertex $v$ of $P$, a univalent $f^{nq}$-pullback $Q^v \subset P$ and state that this deeper family is *pairwise disjoint* [KL08, Lemma 2.1]. These pieces are components of the domain of $(f^q|_{Q_0\cup Q_1})^n$, hence lie in $Q_0 \cup Q_1$ and remain disjoint from $K_\epsilon \Subset V^0$. Choosing two $Q^v$ and setting $g = f^{nq}$ therefore preserves (H1), (H2), and (H4). Every abstract theorem below remains unchanged; each transition time $q$ becomes $nq$, and the root arrow becomes $f^{nq}(K_\epsilon) = \mathcal{K}_{nq \bmod p}$.

From Section 3 onward, the logical input is only Standing Hypotheses 2.2. Neither (2.5) nor a particular reading of the Kahn–Lyubich construction enters any later proof. Thus the source realization and the abstract package can be audited independently.

*Remark* 2.7 (Where each hypothesis is used). (H1) and (H4) drive every separation statement (Lemma 5.1). (H2) is used three ways: to push the nested intersection through $g$ (Theorem 5.3), to transport fullness and Fill by the pointed Schoenflies extension (Lemma 3.3 and Theorem 6.1), and to transport cells, ports, and prime-end chains (Theorems 6.4 and 7.2). (H3) is used exactly once, to upgrade $\mathcal{H}_\partial \subset K(f)$ to $\mathcal{H}_\partial \subset J(f)$ (Theorem 5.3); every other statement is independent of it.

## 3. Plane-topology toolkit

For a compact set $K \subset \mathbb{C}$ let

$$\operatorname{Fill}(K) = K \cup \bigcup \{\text{bounded components of } \mathbb{C} \setminus K\}. \tag{3.1}$$

A compact connected set $K \subsetneq S^2$ is *full* if $S^2 \setminus K$ is connected. For bounded $K \subset \mathbb{C}$, viewing $\mathbb{C} \subset \widehat{\mathbb{C}} = S^2$, this is equivalent to $\operatorname{Fill}(K) = K$; the spherical formulation is the one used below, since the packets will include the exterior continuum $S^2 \setminus U$ of a Jordan domain.

The first lemma is the Čech form of Alexander duality. Hatcher explains the passage from the locally contractible case to arbitrary compacta [Hat02, pp. 256–257]; the underlying Čech theory is developed in Eilenberg–Steenrod [ES52]. For every nonempty proper compact $K \subset S^2$,

$$\widetilde{H}_0(S^2 \setminus K; \mathbb{Z}) \;\cong\; \check{H}^1(K; \mathbb{Z}). \tag{3.2}$$

The Čech form, rather than the locally contractible special case, is required: the compacta below are Julia-set pullbacks with no local structure assumptions.

**Lemma 3.1** (Finite disjoint full packets)**.** *If $K_0, \dots, K_N \subset S^2$ are pairwise disjoint nonempty full continua, then $S^2 \setminus \bigcup_{r=0}^N K_r$ is connected.*

*Proof.* Fullness of $K_r$ makes $S^2 \setminus K_r$ connected, so (3.2) gives $\check{H}^1(K_r;\mathbb{Z}) = 0$. The finitely many pairwise disjoint compacta have pairwise disjoint open neighborhoods; the covers of $\bigcup_r K_r$ subordinate to these neighborhoods are cofinal among all open covers, and each has nerve equal to the disjoint union of the individual nerves. Passing to first cohomology and then to the direct limit defining Čech cohomology,

$$\check{H}^1\Big(\bigcup_r K_r;\mathbb{Z}\Big) \cong \bigoplus_r \check{H}^1(K_r;\mathbb{Z}) = 0.$$

A second application of (3.2) makes the complement connected. □

**Lemma 3.2** (Nested finite-full-packet nonseparation)**.** *Let $F_0 \supset F_1 \supset \cdots$ be compact proper subsets of $S^2$, each a finite union of pairwise disjoint full continua. Then $\bigcap_m F_m$ has connected complement in $S^2$. If moreover $U$ is a Jordan domain with $F_m \Subset U$ for every $m$ and each $F_m \cup (\widehat{\mathbb{C}} \setminus U)$ is again a finite union of pairwise disjoint full continua, then $U \setminus \bigcap_m F_m$ is connected.*

*Proof.* By Lemma 3.1 every $S^2 \setminus F_m$ is connected and nonempty. These complements increase with $m$, all contain a common point (any point of $S^2 \setminus F_0$), and their union is the complement of the intersection; an increasing union of connected sets with a common point is connected. For the localized statement, apply the same argument to the packets $F_m \cup (\widehat{\mathbb{C}} \setminus U)$: the complement of the intersection is then exactly $U \setminus \bigcap_m F_m$. □

**Lemma 3.3** (Pointed Schoenflies transport)**.** *Let $\phi\colon D \to D'$ be an orientation-preserving homeomorphism of closed Jordan disks in $\mathbb{C}$. Then $\phi$ extends to an orientation-preserving homeomorphism $\Phi$ of $S^2 = \widehat{\mathbb{C}}$ with $\Phi(\infty) = \infty$. Consequently, for every compact $S \subset D$:*

(a) *if $S$ is a full continuum, so is $\phi(S)$; and*
(b) $\Phi(\mathrm{Fill}(S)) = \mathrm{Fill}(\phi(S))$*; in particular* $\mathrm{Fill}(\phi(S)) = \phi(\mathrm{Fill}(S))$ *whenever* $\mathrm{Fill}(S) \subset D$.

*Proof.* The complements $\Delta = S^2 \setminus \operatorname{int} D$ and $\Delta' = S^2 \setminus \operatorname{int} D'$ are closed Jordan disks in $S^2$ containing $\infty$ in their interiors. By the Schoenflies theorem choose homeomorphisms $h\colon \Delta \to \overline{\mathbb{D}}$ and $h'\colon \Delta' \to \overline{\mathbb{D}}$; composing with self-homeomorphisms of $\overline{\mathbb{D}}$ that fix the boundary pointwise, arrange $h(\infty) = h'(\infty) = 0$. The circle homeomorphism $\beta = h' \circ (\phi|_{\partial D}) \circ h^{-1}$ extends by coning, $\widehat{\beta}(re^{i\theta}) = r\,\beta(e^{i\theta})$, to a homeomorphism of $\overline{\mathbb{D}}$ fixing 0. Define $\Phi = \phi$ on $D$ and $\Phi = (h')^{-1} \circ \widehat{\beta} \circ h$ on $\Delta$; the two definitions agree on $\partial D$, so $\Phi$ is a sphere homeomorphism, and $\Phi(\infty) = (h')^{-1}(0) = \infty$. It is orientation-preserving because it is so on the open disk $\operatorname{int} D$.

For (b): $\Phi$ carries the components of $\widehat{\mathbb{C}} \setminus S$ bijectively onto the components of $\widehat{\mathbb{C}} \setminus \phi(S)$ and matches the component of $\infty$ with the component of $\infty$; hence it carries the union of $S$ with its bounded complementary components onto the corresponding union for $\phi(S)$. When $\mathrm{Fill}(S) \subset D$, the restriction of $\Phi$ to $\mathrm{Fill}(S)$ is $\phi$ itself, giving the displayed identity with $\phi$. (a) is the special case in which $S$ has no bounded complementary components: then neither does $\phi(S)$, and $S^2 \setminus \phi(S) = \Phi(S^2 \setminus S)$ is connected. □

**Lemma 3.4** (Full-continuum Jordan neighborhood)**.** *If a nondegenerate full continuum $K$ is compactly contained in an open set $G \subset \mathbb{C}$, there is a closed Jordan disk $D$ with*

$$K \subset \operatorname{int} D \subset D \Subset G. \tag{3.3}$$

*Proof.* Uniformize the simply connected exterior by $\psi\colon \{|z| > 1\} \to \widehat{\mathbb{C}} \setminus K$. The filled interiors of the analytic Jordan curves $\psi(\{|z| = R\})$, $R > 1$, are closed Jordan disks decreasing to $K$ as $R \downarrow 1$. If none lay in $G$, compactness would place a point of the closed set $\widehat{\mathbb{C}} \setminus G$ in their intersection $K$, a contradiction. □

**Lemma 3.5** (One-point tree attachments are full)**.** *If a finite embedded plane tree meets each of finitely many pairwise disjoint full continua in exactly one point and otherwise misses them, the union is a full continuum.*

*Proof.* A finite embedded plane tree is nonseparating. Attach the full continua one at a time. At each step the two nonseparating continua meet in a connected singleton, and the Janiszewski separation lemma [Kur68, §61] implies that their union is nonseparating. Induction proves that the final compact connected set has connected complement. (Equivalently: connectedness of each intersection and vanishing of the first Čech cohomology of the two pieces force, by Mayer–Vietoris and (3.2), vanishing first Čech cohomology of the union.) □

**Lemma 3.6** (One-ended slit lemma)**.** *Let $\Omega \subset \mathbb{C}$ be a domain and let $\delta_1, \dots, \delta_s \colon [0,1] \to \overline{\Omega}$ be pairwise disjoint Jordan arcs with $\delta_i(0) \in \partial\Omega$ and $\delta_i((0,1]) \subset \Omega$. Then*

$$\Omega \setminus \bigcup_{i=1}^{s} \delta_i((0,1])$$

*is path connected. Moreover,*

$$\Omega \setminus \bigcup_{i=1}^{s} \delta_i((0,1))$$

*is path connected and contains every free tip $\delta_i(1)$.*

*Proof.* For the first statement, treat one slit $\delta$ first. A polygonal path between two points of the complement is compactly contained in $\Omega$, hence misses an initial tail $\delta((0,\varepsilon))$. Perturb the path to meet the compact arc $\delta([\varepsilon,1])$ transversely in finitely many points. In a thin topological rectangle around that arc, replace every crossing by travel along the two long sides; the sides join around the free endpoint $\delta(1)$. (Planar arcs are tame [Moi77, Chap. 10, Theorem 10.8], so such rectangle neighborhoods exist; the same fact underlies every detour argument of this kind.) The replacement stays in $\Omega$ and avoids the whole slit. The removed slit is relatively closed in $\Omega$, so its complement is again a domain, and induction over the finite disjoint family proves the claim.

For the second statement, write $O = \Omega \setminus \bigcup_i \delta_i((0,1])$, which is open, path connected by the first part, and misses every tip. Fix a tip $p_i = \delta_i(1)$. By injectivity and compactness there is $t_i < 1$ with $\operatorname{dist}(p_i, \delta_i([0,t_i])) > 0$; choose an open disk $B_i \ni p_i$ with $\overline{B_i} \subset \Omega$, meeting no $\delta_j$ with $j \neq i$, missing $\delta_i([0,t_i])$, and with the $B_i$ pairwise disjoint. By tameness of the arc $A_i = \delta_i([t_i,1])$ [Moi77, Chap. 10, Theorem 10.8], a homeomorphism $h$ of the plane carries $A_i$ to the segment $[0,1]\times\{0\}$ with $h(p_i) = (1,0)$. For small $s > 0$ the segment $\{(1+\sigma, 0) : 0 \le \sigma \le s\}$ lies in the open set $h(B_i)$ and, except for its initial point, misses $h(A_i)$; its preimage is an arc from $p_i$ into $B_i$ whose interior misses every slit, hence lies in $O$. Thus every tip is joined to the path-connected set $O$ by an arc meeting the slits only at the tip itself, and $O \cup \{p_1, \dots, p_s\}$ is path connected. □

3.1. **Crosscut chains.** Finally we fix the standard crosscut and prime-end conventions; see [Pom92, §2.4]. A *crosscut* of the complement of a full continuum $A$ is a Jordan arc whose interior lies in $\widehat{\mathbb{C}} \setminus A$ and whose two distinct endpoints lie on $\partial A$. A *crosscut chain* is a sequence

$$(\gamma_n, D_n)_{n \ge 1}$$

of pairwise disjoint crosscuts $\gamma_n$ with selected sides $D_n$, where:

(C1) $D_n$ is a selected component of $\widehat{\mathbb{C}} \setminus (A \cup \gamma_n)$;

(C2) the chain is *nulled*:

$$\overline{D_{n+1}} \setminus A \subset D_n \qquad (n \geq 1); \tag{3.4}$$

(C3) $\operatorname{diam} \gamma_n \to 0$.

Condition (3.4) contains the usual null-chain relation: since the open arc $\gamma_{n+1}^{\circ}$ lies in $\overline{D_{n+1}} \setminus A$, the later crosscuts satisfy $\gamma_{n+1}^{\circ} \subset D_n$, so for $n \geq 2$ every crosscut interior lies in $D_1$. The chain defines a prime end of $\widehat{\mathbb{C}} \setminus A$ with impression $\bigcap_n \overline{D_n}$. We call $\overline{D_n} \setminus A$ the *relative collar tail* at level $n$. Relative collar tails of different chains may have closures meeting inside $A$—in particular, overlapping prime-end impressions are allowed, and no local connectivity of $A$ is ever assumed.

**Lemma 3.7** (Protected crosscut survival and order)**.** *Let $A$ be a full continuum, let $H \supset A$ be a full continuum, and let $(\gamma_n, D_n)$ be a crosscut chain of $\widehat{\mathbb{C}} \setminus A$. If*

$$(H \setminus A) \cap (\overline{D_1} \setminus A) = \varnothing, \tag{3.5}$$

*then the crosscuts $(\gamma_n)_{n\geq 2}$ with their selected sides form a prime-end chain in $\widehat{\mathbb{C}} \setminus H$; the endpoints remain boundary points and the selected sides are the same domains $D_n$. A finite family of such chains with pairwise disjoint relative terminal collar tails induces distinct prime ends of $\widehat{\mathbb{C}} \setminus H$ with unchanged cyclic order. Here the preserved cyclic order is that of the transported crosscut family in the one ambient complement $\widehat{\mathbb{C}} \setminus H$; no identification of prime-end circles across different atoms is asserted.*

*Proof.* For $n \geq 2$, (3.4) gives $\gamma_n^{\circ} \subset D_{n-1} \subset D_1$, so $\gamma_n^{\circ} \subset \overline{D_1} \setminus A$ misses $H$ by (3.5), while the two distinct endpoints of $\gamma_n$ lie in $A \subset H$. Every neighborhood of either endpoint meets the crosscut interior, so both endpoints remain on the boundary of the enlarged complement; thus $\gamma_n$ remains a crosscut of $\widehat{\mathbb{C}} \setminus H$.

Moreover, $D_n \subset \overline{D_n} \setminus A \subset D_1$ for $n \geq 2$, so $D_n \cap H = \varnothing$. As a component of $\widehat{\mathbb{C}} \setminus (A \cup \gamma_n)$, its boundary satisfies $\partial D_n \subset A \cup \gamma_n \subset H \cup \gamma_n$. Hence $D_n$ itself is a component of $\widehat{\mathbb{C}} \setminus (H \cup \gamma_n)$: it is connected and both open and closed relative to that complement. Therefore the selected sides, their nesting relation, and the crosscut diameters are unchanged.

Equivalent chains eventually divide one another; pairwise disjoint relative terminal collar tails prevent that, so the induced prime ends are distinct. For a finite family, choose one deep oriented crosscut from each chain: prime-end cyclic order is the separation order of these crosscuts, and neither the crosscuts nor their selected sides changed, so every ordered triple retains its orientation. □

## 4. Protected regulated connectors

This section constructs the connector technology consumed by the protected-carrier theorem in Section 7. Throughout, $U$ is a Jordan domain, $A_1, \dots, A_s \Subset U$ are pairwise disjoint full continua (the *atoms*), and $Z \Subset U \setminus \bigcup_i A_i$ is a compact *forbidden set*. For every protected access to $A_i$ a crosscut chain $\Gamma_{i,a} = (\gamma_{i,a,n}, D_{i,a,n})_{n\geq 1}$ of $\widehat{\mathbb{C}} \setminus A_i$ is chosen, truncated so that the relative terminal collar tails $\overline{D_{i,a,1}} \setminus A_i$ are pairwise disjoint, lie in $U$, and miss $Z$ and the other atoms. The closures of two tails may meet inside their common atom: overlapping prime-end impressions are allowed.

**Definition 4.1** (Protected ribbon-routing certificate)**.** A *protected ribbon-routing certificate* consists of the preceding data, pairwise disjoint *free-port arcs* $\alpha_i$ satisfying

$$\alpha_i \cap A_i = \{x_i\} = \{\alpha_i(0)\}, \qquad \alpha_i((0,1]) \cap \Big( Z \cup \bigcup_j A_j \cup \bigcup_{j,a} (\overline{D_{j,a,1}} \setminus A_j) \Big) = \varnothing,$$

and an embedded finite tree $G \subset U$ joining the outer port endpoints $\alpha_i(1)$ while missing the same protected and forbidden sets, subject to the incidence conditions

$$G \cap \alpha_i = \{\alpha_i(1)\} \quad (1 \le i \le s), \qquad T_{\mathfrak{R}} := G \cup \bigcup_i \alpha_i \text{ is a finite embedded plane tree.} \tag{4.1}$$

The planar cyclic order of half-edges at every vertex of $T_{\mathfrak{R}}$ is retained. The free port points $x_i$ are required not to be impressions of the protected chains.

Given a certificate $\mathfrak{R}$, define the *carrier*

$$H_{\mathfrak{R}} = T_{\mathfrak{R}} \cup \bigcup_i A_i = G \cup \bigcup_i \alpha_i \cup \bigcup_i A_i. \tag{4.2}$$

The *attachment trace* of a carrier $H \supset A_i$ at the atom $A_i$ is the compact set $\overline{H \setminus A_i} \cap A_i$.

**Theorem 4.2** (Protected regulated connector realization)**.** *Every protected ribbon-routing certificate has an actual carrier $H_{\mathfrak{R}}$ with the following properties:*

(i) *$H_{\mathfrak{R}} \Subset U$ is compact, connected, and full;*
(ii) *it contains every $A_i$, misses $Z$, and has singleton attachment trace: $\overline{H_{\mathfrak{R}} \setminus A_i} \cap A_i = \{x_i\}$ for every $i$;*
(iii) *every protected crosscut chain remains a prime-end chain in $\widehat{\mathbb{C}} \setminus H_{\mathfrak{R}}$;*
(iv) *the survival map on the protected prime ends is injective and preserves the cyclic order of the transported family in $\widehat{\mathbb{C}} \setminus H_{\mathfrak{R}}$;*
(v) *the connector-created sides carry the boundary order of the retained ribbon tree; and*
(vi) *no local connectivity of any $A_i$ is required.*

*Proof.* By (4.1), $T_{\mathfrak{R}}$ is a finite embedded plane tree, and it meets each atom $A_i$ exactly in the singleton $\{x_i\}$: the port $\alpha_i$ meets $A_i$ exactly there, the other ports miss $A_i$ by the certificate, and $G$ misses every atom. Lemma 3.5 applied to $T_{\mathfrak{R}}$ and the atoms gives (i) for the union (4.2), and the certificate gives the containment and avoidance parts of (ii). For the attachment trace, $H_{\mathfrak{R}} \setminus A_i$ is the union of $T_{\mathfrak{R}} \setminus \{x_i\}$ with the other atoms; its closure adds only $x_i$, and the other atoms are closed and disjoint from $A_i$, so $\overline{H_{\mathfrak{R}} \setminus A_i} \cap A_i = \{x_i\}$.

For a protected chain at $A_i$, the set $H_{\mathfrak{R}} \setminus A_i$ misses the relative tail $\overline{D_{i,a,1}} \setminus A_i$ by the certificate, verifying (3.5); Lemma 3.7 then proves (iii) and (iv), including the stated reading of cyclic order. A small regular neighborhood of the finite tree $T_{\mathfrak{R}}$ traverses every edge on two sides in exactly the cyclic boundary word of its ribbon structure, giving (v). The argument uses crosscut chains only, so no Euclidean boundary extension of an atom is assumed, giving (vi). □

The existence of a certificate is characterized by a free-component condition. Freeze the inventory: let $Q$ be the union of all atoms, all relative terminal collar tails $\overline{D_{i,a,1}} \setminus A_i$, and the forbidden set $Z$, and put

$$\Omega_{\text{free}} = U \setminus Q. \tag{4.3}$$

This $Q$ is compact. Although a relative tail need not be closed by itself,

$$A_i \cup \left(\overline{D_{i,a,1}} \setminus A_i\right) = A_i \cup \overline{D_{i,a,1}}$$

is compact, and the inventory is finite. Consequently $\Omega_{\text{free}}$ is an open plane domain. A component $\Omega_0$ of $\Omega_{\text{free}}$ has *simultaneous singleton ports* if there are pairwise disjoint arcs $\alpha_i$ with $\alpha_i \cap A_i = \{\alpha_i(0)\} = \{x_i\}$, $x_i$ not an impression of a protected chain, and $\alpha_i((0,1]) \subset \Omega_0$.

**Theorem 4.3** (Protected free-component criterion)**.** *A protected ribbon-routing certificate exists if and only if one component of $\Omega_{\text{free}}$ has simultaneous singleton ports to every required atom.*

*Proof.* Necessity: a certificate's tree $T_{\mathfrak{R}}$ with its atoms deleted is connected and lies in $\Omega_{\text{free}}$, so all its port interiors lie in one component.

Sufficiency: choose the simultaneous port arcs in $\Omega_0$. Their interiors $\alpha_i((0,1])$ are finitely many disjoint one-ended slits of the domain $\Omega_0$, with $\alpha_i(0) = x_i \in \partial\Omega_0$. By the Moreover clause of Lemma 3.6, the set

$$O^+ = \Omega_0 \setminus \bigcup_i \alpha_i((0,1))$$

is path connected and contains every outer tip $\alpha_i(1)$. Join the finitely many tips pairwise by paths in $O^+$; since $O^+ \setminus \{\text{tips}\} = \Omega_0 \setminus \bigcup_i \alpha_i((0,1])$ is open, the path interiors may be perturbed within that open set into general position, avoiding the finitely many tips except at their own endpoints, and then subdivided at their finitely many crossings. An inclusion-minimal connected subgraph $G$ spanning the tip vertex set is a finite embedded tree with $G \cap \alpha_i = \{\alpha_i(1)\}$ and $G \cap \bigcup_i \alpha_i((0,1)) = \varnothing$. Attaching each port arc $\alpha_i$ to $G$ at the single point $\alpha_i(1)$—the ports being pairwise disjoint and meeting $G$ nowhere else—yields a finite embedded plane tree $T_{\mathfrak{R}} = G \cup \bigcup_i \alpha_i$, so (4.1) holds and the data form a protected ribbon-routing certificate. □

## 5. The rooted-end compactification

5.1. **Addresses and fibers.** Let $\mathtt{A} = \{0,1\}$, let $\mathtt{A}^{<\mathbb{N}}$ denote the finite binary words including the empty word $\epsilon$, and let $\mathtt{A}^{\mathbb{N}} = \mathtt{A}^{\mathbb{N}}$ denote the ends. For $w = a_1 \cdots a_m \in \mathtt{A}^{<\mathbb{N}}$ put

$$\phi_w = \phi_{a_1} \circ \cdots \circ \phi_{a_m}, \qquad Q_w = \phi_w(P), \qquad K_w = \phi_w(K_\epsilon), \tag{5.1}$$

with $\phi_\epsilon = \mathrm{id}$, $Q_\epsilon = P$, $K_\epsilon = K_\epsilon$. For $\omega \in \mathtt{A}^{\mathbb{N}}$ put

$$E_\omega = \bigcap_{m \geq 1} Q_{\omega|m}. \tag{5.2}$$

The *binary address compactification* is

$$\Omega_\partial = \{w\#^\infty : w \in \mathtt{A}^{<\mathbb{N}}\} \cup \mathtt{A}^{\mathbb{N}} \ \subset\ \{0,1,\#\}^{\mathbb{N}}, \tag{5.3}$$

with the product topology; we identify $w$ with $w\#^\infty$. The *rooted-end compactum* is

$$\mathcal{H}_\partial = \bigcup_{w \in \mathtt{A}^{<\mathbb{N}}} K_w \ \cup \ \bigcup_{\omega \in \mathtt{A}^{\mathbb{N}}} E_\omega. \tag{5.4}$$

Write $\mathrm{fib}(w) = K_w$ and $\mathrm{fib}(\omega) = E_\omega$ for the realization map on $\Omega_\partial$.

Two words are *comparable* if one is a prefix of the other (written $u \sqsubseteq v$), and an end $\omega$ *extends* $u$ if $\omega|_{|u|} = u$.

**Lemma 5.1** (Cylinder separations)**.** *Under Standing Hypotheses 2.2:*

(a) *$Q_{wa} = \phi_w(Q_a) \subset Q_w$; appending symbols descends through nested closed Jordan disks, and $g^{|w|}\colon Q_w \to P$ is a homeomorphism.*
(b) *If $u, v$ are incomparable, then $Q_u \cap Q_v = \varnothing$.*
(c) *If $u$ is a strict prefix of $v$, then $K_u \cap Q_v = \varnothing$. In particular $K_u$ is disjoint from $K_v$ and from every $E_\omega$ with $\omega$ extending $u$.*
(d) *Distinct fibers of the realization map are pairwise disjoint.*

*Proof.* (a) is immediate from (H2) and $\phi_{wa} = \phi_w \circ \phi_a$. For (b), write $u = cas$, $v = ca't$ with $a \neq a'$ for the longest common prefix $c$; then $Q_u \subset Q_{ca} = \phi_c(Q_a)$ and $Q_v \subset \phi_c(Q_{a'})$, and these are disjoint because $Q_a \cap Q_{a'} = \varnothing$ by (H1) and $\phi_c$ is injective. For (c), write $v = uat$; then $Q_v \subset \phi_u(Q_a)$ while $K_u = \phi_u(K_\epsilon)$ with $K_\epsilon \cap Q_a = \varnothing$ by (H4), and injectivity of $\phi_u$ separates them. (d) combines (b) and (c): incomparable addresses have disjoint cylinders at the first difference, and comparable distinct addresses fall under (c) (every end fiber $E_\omega$ lies in $Q_{\omega|m}$ for all $m$, and two distinct ends are incomparable at some depth). □

**Lemma 5.2** (Label-limit lemma)**.** *Let $\alpha_j \to \alpha$ in $\Omega_\partial$ and let $x_j \in \mathrm{fib}(\alpha_j)$ with $x_j \to x$ in $\mathbb{C}$. Then $x \in \mathrm{fib}(\alpha)$.*

*Proof.* If $\alpha = w\#^\infty$ is a finite address, it is isolated in $\Omega_\partial$: any address agreeing with it through position $|w|+1$ (that is, on $w$ followed by $\#$) equals it. Hence $\alpha_j = \alpha$ eventually, and $x \in K_w$ because $K_w$ is closed. If $\alpha = \omega \in \mathtt{A}^{\mathbb{N}}$, fix $m$. Eventually $\alpha_j$ agrees with $\omega$ on the first $m$ symbols; whether $\alpha_j$ is a finite word $v \sqsupseteq \omega|m$ or an end extending $\omega|m$, its fiber lies in the closed cylinder $Q_{\omega|m}$ by Lemma 5.1(a). Hence $x \in Q_{\omega|m}$ for every $m$, that is, $x \in E_\omega$. □

## 5.2. **The compactification theorem.**

**Theorem 5.3** (Direct-decoration rooted-end compactification)**.** *Under Standing Hypotheses 2.2:*

(i) *$\Omega_\partial$ is compact; the finite addresses are isolated, and finite words of increasing length converge to an end exactly by eventual agreement on every fixed prefix.*

(ii) *$\mathcal{H}_\partial$ is a compact subset of $K(f)$; under (H3), $\mathcal{H}_\partial \subset J(f)$. Every fiber (5.1), (5.2) is a nonempty continuum.*

(iii) *The connected components of $\mathcal{H}_\partial$ are exactly the fibers $K_w$ ($w \in \mathtt{A}^{<\mathbb{N}}$) and $E_\omega$ ($\omega \in \mathtt{A}^{\mathbb{N}}$).*

(iv) *The realization $\Omega_\partial \ni \alpha \mapsto \mathrm{fib}(\alpha)$ has closed graph; equivalently, it is upper semicontinuous as a map into the compact subsets of $\mathcal{H}_\partial$.*

(v) *The dynamics acts through actual univalent branches of $f^q$:*

$$g(K_{aw}) = K_w, \qquad g(E_{a\omega}) = E_\omega \qquad (a \in \mathtt{A}), \tag{5.5}$$

*each transition having time $q$ and inverse sheet the deleted first symbol $a$.*

*Proof.* (i). The product $\{0,1,\#\}^{\mathbb{N}}$ is compact. The subset (5.3) is defined by the closed conditions "if the $k$-th symbol is $\#$ then so is the $(k+1)$-st": each violating set is a clopen cylinder, so $\Omega_\partial$ is closed, hence compact. Isolation of finite addresses and the stated convergence are immediate in the product topology (see the proof of Lemma 5.2).

(ii). Each $Q_w$ is a closed Jordan disk and $g^{|w|}\colon Q_w \to P$ is a homeomorphism (Lemma 5.1(a)), so each $K_w = \phi_w(K_\epsilon)$ is a continuum, and each $E_\omega$ is a nested intersection of continua, hence a nonempty continuum. Every $z \in E_\omega$ satisfies $g^m(z) \in P$ for all $m \geq 0$; the intermediate iterates $f^{mq+r}(z)$, $0 \leq r < q$, lie in the compact union $\bigcup_{r=0}^{q-1} f^r(P)$, so the forward orbit of $z$ is bounded and $z \in K(f)$. The root satisfies $K_\epsilon \subset K(f)$: its points have their $f^p$-orbits in $K_\epsilon$. For $z \in K_\epsilon$, write an iterate as $mp+i$ with $0 \leq i < p$; then

$$f^{mp+i}(z) \in f^i(K_\epsilon) = \mathcal{K}_i.$$

The finite union $\bigcup_{i=0}^{p-1} \mathcal{K}_i$ is compact, so the full forward orbit of $z$ is bounded. Complete invariance of $K(f)$ then puts every $K_w = \phi_w(K_\epsilon) \subset f^{-q|w|}(K(f)) = K(f)$. Under (H3), $K(f) = J(f)$.

For compactness of $\mathcal{H}_\partial$, let $x_j \in \mathcal{H}_\partial$ converge to $x$, and let $\alpha_j \in \Omega_\partial$ be addresses with $x_j \in \mathrm{fib}(\alpha_j)$. By compactness of $\Omega_\partial$ pass to a subsequence with $\alpha_j \to \alpha$; Lemma 5.2 gives $x \in \mathrm{fib}(\alpha) \subset \mathcal{H}_\partial$.

(iv). The same argument applied without passing to a subsequence of the $x_j$ proves that the graph $\{(\alpha, x) : x \in \mathrm{fib}(\alpha)\}$ is closed in $\Omega_\partial \times \mathcal{H}_\partial$. Since $\mathcal{H}_\partial$ is compact, a closed graph is equivalent to upper semicontinuity of the set-valued realization.

(iii). Each fiber is connected by (ii), and distinct fibers are disjoint by Lemma 5.1(d). It remains to show that no connected subset of $\mathcal{H}_\partial$ meets two fibers. For a finite word $u$, consider

$$\mathcal{H}_\partial[u] := \bigcup_{v \sqsupseteq u} K_v \cup \bigcup_{\omega \sqsupseteq u} E_\omega.$$

We claim $\mathcal{H}_\partial[u]$ is clopen in $\mathcal{H}_\partial$. First, $\mathcal{H}_\partial[u] = \mathcal{H}_\partial \cap Q_u$: fibers with address extending $u$ lie in $Q_u$ by Lemma 5.1(a), fibers with address incomparable to $u$ miss $Q_u$ by (b), and fibers $K_v$ with $v$ a strict prefix of $u$ miss $Q_u$ by (c). Hence $\mathcal{H}_\partial[u]$ is closed. Second, its complement $\mathcal{H}_\partial \setminus \mathcal{H}_\partial[u]$ is the union of the fibers over the addresses not extending $u$; these addresses form a clopen, hence compact, subset $N_u \subset \Omega_\partial$, and the union of fibers over any compact address set is closed: the graph of the realization restricted to $N_u$ is a closed subset of the compact space $N_u \times \mathcal{H}_\partial$ ((iv)), hence compact, and its projection to $\mathcal{H}_\partial$ is therefore compact. Thus $\mathcal{H}_\partial[u]$ is clopen.

Now let $C \subset \mathcal{H}_\partial$ be connected and meet the fibers of two distinct addresses $\alpha \neq \beta$. If $\alpha, \beta$ first differ at position $k$ (both defined and unequal there), the clopen sets $\mathcal{H}_\partial[\alpha|_k]$ and its complement separate the two fibers, so $C$ cannot meet both. Otherwise one address is a finite word $u$ and the other strictly extends it; then $\beta$ extends $ua$ for some $a \in \mathtt{A}$, the fiber of $\beta$ lies in the clopen set $\mathcal{H}_\partial[ua]$, and $K_u$ lies in its complement by Lemma 5.1(c). In both cases $C$ is disconnected by a clopen partition, a contradiction. So the components of $\mathcal{H}_\partial$ are exactly the fibers.

(v). From $g \circ \phi_a = \mathrm{id}_P$ we get $g \circ \phi_{aw} = \phi_w$, hence $g(K_{aw}) = K_w$. Since $g \colon Q_a \to P$ is a homeomorphism (H2), it commutes with the nested intersection (5.2):

$$g(E_{a\omega}) = g\Big(\bigcap_m Q_{(a\omega)|m}\Big) = \bigcap_m g\big(Q_{a(\omega|m-1)}\big) = \bigcap_m Q_{\omega|m-1} = E_\omega.$$

Each transition is the restriction of $f^q$ to a subset of $Q_a$, and the deleted first symbol $a$ records the inverse sheet. □

### 5.3. **The component quotient and the exact partial shift.**

**Theorem 5.4** (Component quotient)**.** *Define $\pi_\partial \colon \mathcal{H}_\partial \to \Omega_\partial$ by $\pi_\partial(x) = \alpha \iff x \in \mathrm{fib}(\alpha)$. Then $\pi_\partial$ is a well-defined continuous closed surjection whose fibers are precisely the connected components of $\mathcal{H}_\partial$. Consequently the component space of $\mathcal{H}_\partial$, with the quotient topology, is homeomorphic to $\Omega_\partial$:*

$$\mathcal{H}_\partial / \sim_{\mathit{component}} \;\cong\; \Omega_\partial.$$

*In particular the component decomposition of $\mathcal{H}_\partial$ is upper semicontinuous.*

*Proof.* Well-definedness and surjectivity hold because the nonempty fibers are pairwise disjoint and cover $\mathcal{H}_\partial$ (Lemma 5.1(d), Theorem 5.3(ii)). For continuity, let $x_j \to x$ in $\mathcal{H}_\partial$ and write $\alpha_j = \pi_\partial(x_j)$, $\alpha = \pi_\partial(x)$. If $\alpha_j \not\to \alpha$, choose a neighborhood $V$ of $\alpha$ that infinitely many $\alpha_j$ avoid; compactness of $\Omega_\partial \setminus V$ gives a subsequence converging to some $\beta \neq \alpha$. Lemma 5.2 then puts $x \in \mathrm{fib}(\beta)$, while $x \in \mathrm{fib}(\alpha)$, contradicting disjointness of fibers. Hence $\alpha_j \to \alpha$ and $\pi_\partial$ is continuous. A continuous map from a compact space to a Hausdorff space is closed; a closed continuous surjection is a quotient map, and its fibers are the components by Theorem 5.3(iii). Upper semicontinuity of the decomposition is exactly closedness of the quotient map with compact fibers. □

**Theorem 5.5** (Exact partial shift realization)**.** *For $a \in \mathtt{A}$ put $H_a = \mathcal{H}_\partial \cap Q_a$. Then:*

(i) $H_a = \phi_a(\mathcal{H}_\partial)$, *and* $H_0, H_1$ *are disjoint clopen subsets of* $\mathcal{H}_\partial$ *with*

$$\mathcal{H}_\partial \setminus K_\epsilon = H_0 \sqcup H_1;$$

(ii) $g|_{H_a} \colon H_a \to \mathcal{H}_\partial$ *is a homeomorphism with inverse* $\phi_a$*; hence*

$$g \colon \mathcal{H}_\partial \setminus K_\epsilon \longrightarrow \mathcal{H}_\partial$$

*is a (trivial) two-sheeted covering with clopen sheets* $H_0, H_1$*;*

(iii) *the component quotient semiconjugates this covering to the first-symbol deletion shift:*

$$\pi_\partial \circ g = \sigma_{\mathrm{pre}} \circ \pi_\partial \qquad \text{on } \mathcal{H}_\partial \setminus K_\epsilon.$$

*Proof.* (i) The identities $\phi_a(K_w) = K_{aw}$ and $\phi_a(E_\omega) = E_{a\omega}$ show $\phi_a(\mathcal{H}_\partial) = \bigcup_w K_{aw} \cup \bigcup_\omega E_{a\omega} = \mathcal{H}_\partial[a]$, which equals $\mathcal{H}_\partial \cap Q_a$ and is clopen by the proof of Theorem 5.3(iii). Every fiber other than the root fiber has an address beginning with 0 or 1, so $\mathcal{H}_\partial \setminus K_\epsilon = H_0 \sqcup H_1$, and $H_0 \cap H_1 = \varnothing$ by (H1).

(ii) On $H_a \subset Q_a$ the map $g$ is the restriction of the homeomorphism $g \colon Q_a \to P$ of (H2), with inverse $\phi_a$; by (i), $\phi_a$ carries $\mathcal{H}_\partial$ onto $H_a$, so $g|_{H_a}$ is a homeomorphism onto $\mathcal{H}_\partial$. The two sheets are clopen and disjoint, so the restriction of $g$ to their union is a trivial two-sheeted covering of $\mathcal{H}_\partial$.

(iii) On $H_a$, the fiber of address $a\beta$ maps onto the fiber of address $\beta$ by (5.5); this is exactly the semiconjugacy identity. □

5.4. **The two parent maps and the root arrow.** The two arrows are easiest to compare when their geometries are kept in separate panels.

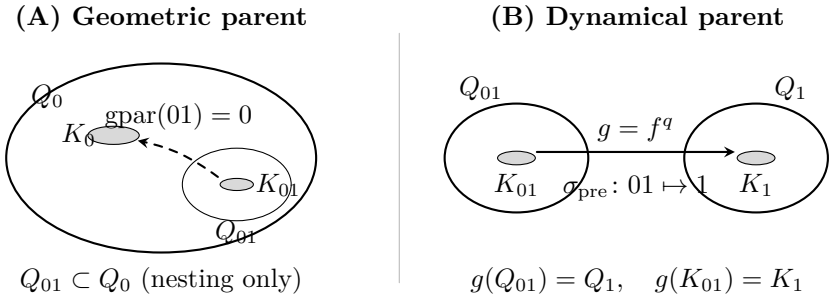



FIGURE 1. The two parent maps at address 01, separated by type. (A) Last-symbol deletion records the inclusion $Q_{01} = \phi_0(Q_1) \subset Q_0$ and has no dynamical content. (B) First-symbol deletion is the actual $f^q$-transition. Address fibers use word subscripts $K_w$; terminal-cycle components use the distinct notation $\mathcal{K}_i$.

*Remark* 5.6 (Two parent maps). The package carries two structurally different parent arrows (Figure 1):

$$\mathrm{gpar}(wa) = w, \qquad \sigma_{\mathrm{pre}}(a\alpha) = \alpha. \tag{5.6}$$

The *geometric parent* gpar deletes the *last* symbol of a nonempty finite word. It is realized by cylinder inclusion, $Q_{wa} \subset Q_w$, and by *no* dynamical map: there is in general no iterate of $f$ carrying $K_{wa}$ onto $K_w$. It has no action on ends, which have no last symbol.

The *dynamical parent* $\sigma_{\mathrm{pre}}$ deletes the *first* symbol. It is defined on all non-root finite addresses and on *all* ends, and it is realized by the exact two-branch covering of Theorem 5.5, with transition time $q$ and the deleted symbol as inverse sheet. Concretely, at address 01: $\mathrm{gpar}(01) = 0$ because $Q_{01} = \phi_0(Q_1) \subset Q_0$, while $\sigma_{\mathrm{pre}}(01) = 1$ because $g(K_{01}) = K_1$. The

two maps are not equal. On a word of length at least two they agree precisely for constant words; for example, they agree on 00 but differ on 01. Conflating them collapses the distinction between nesting depth and dynamical time, so both are retained as separate typed data.

*Remark* 5.7 (The root arrow is separate data). The dynamical parent $\sigma_{\mathrm{pre}}$ is undefined at the empty word. The root has its own transition, coming from the terminal cycle of Proposition 2.1:

$$\tau_{\mathrm{root}}\colon \quad g(K_\epsilon) = f^q(K_\epsilon) = \mathcal{K}_{q \bmod p}. \tag{5.7}$$

This arrow lands on a component of the terminal cycle; its target is indexed by the cycle rather than by an address, and no positional claim about $\mathcal{K}_{q \bmod p}$ relative to $\mathcal{H}_\partial$ is made beyond Proposition 2.1. It is *not* an extension of $\sigma_{\mathrm{pre}}$, and no assertion is made that the image escapes $\mathcal{H}_\partial$ permanently (Non-claims 2.3(N1)). Keeping $\tau_{\mathrm{root}}$ as a separate typed arrow, rather than force-fitting it into the shift, is precisely what makes the package close up without a circular hypothesis.

5.5. **Naturality under root enlargement.** The virtual program enlarges its root package. The compactification is inclusion-natural under exactly the enlargements that stay in the gap. For a compact connected $R$ with $K_\epsilon \subset R \subset P \setminus (Q_0 \cup Q_1)$, write

$$\mathcal{H}_\partial(R) = \bigcup_{w \in \mathbf{A}^{<\mathbb{N}}} \phi_w(R) \;\cup\; \bigcup_{\omega \in \mathbf{A}^{\mathbb{N}}} E_\omega$$

for the compactum with root $R$; thus $\mathcal{H}_\partial = \mathcal{H}_\partial(K_\epsilon)$.

**Theorem 5.8** (Root-gap enlargement naturality)**.** *Let*

$$K_\epsilon \subset S \subset S^+ \subset P \setminus (Q_0 \cup Q_1) \tag{5.8}$$

*be compact connected roots. Then for each $R \in \{S, S^+\}$:*

(i) *all separations of Lemma 5.1, the conclusion of Lemma 5.2, and conclusions (i), (iii), (iv), (v) of Theorem 5.3 hold for $\mathcal{H}_\partial(R)$; $\mathcal{H}_\partial(R)$ is compact and every fiber is a nonempty continuum;*

(ii) *the component quotient and partial-shift theorems (Theorems 5.4 and 5.5) hold for $\mathcal{H}_\partial(R)$, with root fiber $R$ in place of $K_\epsilon$;*

(iii) *membership in the filled Julia set is exactly*

$$\mathcal{H}_\partial(R) \subset K(f) \iff R \subset K(f),$$

*and under (H3) the same equivalence holds with $J(f)$;*

(iv) *the inclusions $\phi_w(S) \subset \phi_w(S^+)$ commute with $g$ at every finite word, and the end fibers are unchanged.*

*Proof.* (i)–(ii) The proofs of the listed conclusions use the root only through compactness, connectedness, containment in $P$, and disjointness from $Q_0 \cup Q_1$—the consequences of (2.6) actually invoked there—all of which hold for $S$ and $S^+$ by (5.8). The only step of Theorem 5.3(ii) not covered is the containment of the fibers in $K(f)$, which used $K_\epsilon \subset K(f)$ and is treated separately in (iii). The quotient and partial-shift proofs use only the separations, the label-limit lemma, compactness, and (H1)–(H2), so they persist verbatim.

(iii) If $R \subset K(f)$, complete invariance of $K(f)$ gives $\phi_w(R) \subset f^{-q|w|}(K(f)) = K(f)$, and the end fibers lie in $K(f)$ by the bounded-orbit argument of Theorem 5.3(ii); hence $\mathcal{H}_\partial(R) \subset K(f)$. Conversely $R$ is the root fiber of $\mathcal{H}_\partial(R)$, so $\mathcal{H}_\partial(R) \subset K(f)$ forces $R \subset K(f)$. Under (H3), $K(f) = J(f)$.

(iv) Injectivity of $\phi_w$ gives $\phi_w(S) \subset \phi_w(S^+)$, and $g \circ \phi_a = \mathrm{id}$ gives the commuting squares. The end fibers (5.2) depend only on $P, Q_0, Q_1$. □

*Remark* 5.9 (Three levels of enlargement, and the root arrow). Theorem 5.8 is the *topological* level: it needs only a compact connected root in the gap, and it does not assert that $S$ or $S^+$ is full, contained in $J(f)$, dynamically protected, access-complete, or admissible as an analytic virtual carrier; those properties are separate audits. The *fullness/cell* level requires a full root compactly contained in the open gap component $G_0$ and is treated in Corollary 6.7. The *Julia* level is the equivalence (iii). Finally, the root arrow (5.7) refers to the terminal cycle of the original root $K_\epsilon$; for an enlarged root $R$ the image $g(R)$ carries no terminal-cycle identification, and any dynamical meaning of the enlarged root arrow requires a separate audit.

## 6. Fullness, the end compactum, gap cells, and replacement

### 6.1. Full fibers and connected complement.

**Theorem 6.1** (Full fibers and nonseparation). *Every fiber $K_w$ and $E_\omega$ of Theorem 5.3 is a full continuum, and $\widehat{\mathbb{C}} \setminus \mathcal{H}_\partial$ is connected. Moreover, for every Jordan domain $U$ with $P \Subset U$, the relative complement $U \setminus \mathcal{H}_\partial$ is connected.*

*Proof.* The root $K_\epsilon$ is full [DH85]. Each $\phi_w \colon P \to Q_w$ is an orientation-preserving homeomorphism of closed Jordan disks, so $K_w = \phi_w(K_\epsilon)$ is full by Lemma 3.3(a). For an end,

$$\widehat{\mathbb{C}} \setminus E_\omega = \bigcup_{m \geq 1} \left(\widehat{\mathbb{C}} \setminus Q_{\omega|m}\right)$$

is an increasing union of connected sets containing $\infty$, hence connected; so $E_\omega$ is full.

For $m \geq 0$ define the approximants

$$\mathcal{H}_\partial^{(m)} = \bigcup_{|u|<m} K_u \;\cup\; \bigcup_{|w|=m} Q_w. \tag{6.1}$$

Each term is full, and the terms of $\mathcal{H}_\partial^{(m)}$ are pairwise disjoint: equal-depth cylinders are disjoint (Lemma 5.1(b)), distinct finite fibers are disjoint ((d)), and for $|u| < m = |w|$ the word $u$ is either incomparable to $w$ (disjoint cylinders) or a strict prefix of $w$ (then $K_u \cap Q_w = \varnothing$ by (c)). The sets (6.1) decrease in $m$: each $K_u$ with $|u| = m$ lies in $Q_u$, and each $Q_w$ with $|w| = m+1$ lies in $Q_{w|m}$.

Their intersection is exactly $\mathcal{H}_\partial$. Indeed $\mathcal{H}_\partial \subset \mathcal{H}_\partial^{(m)}$ for every $m$ (a fiber with address of length $\geq m$ or an end fiber lies in its depth-$m$ cylinder). Conversely, let $x \in \bigcap_m \mathcal{H}_\partial^{(m)}$. If $x$ lies in some finite fiber $K_u$ we are done. Otherwise, for every $m$ there is a unique word $w_m$ of length $m$ with $x \in Q_{w_m}$ (uniqueness by disjointness of equal-depth cylinders), and $x \in Q_{w_{m+1}} \subset Q_{w_{m+1}|m}$ forces $w_{m+1}|m = w_m$. The words $w_m$ therefore converge to an end $\omega$ with $x \in \bigcap_m Q_{\omega|m} = E_\omega$.

Lemma 3.2 applied to the decreasing packets (6.1) proves that $\widehat{\mathbb{C}} \setminus \mathcal{H}_\partial$ is connected; its localized clause, applied with the exterior full continuum $\widehat{\mathbb{C}} \setminus U$ adjoined to each packet, proves that $U \setminus \mathcal{H}_\partial$ is connected. □

### 6.2. The end compactum.

**Corollary 6.2** (Cantor-indexed end compactum). *Let*

$$\mathcal{H}_\infty = \bigcup_{\omega \in \mathtt{A}^{\mathbb{N}}} E_\omega.$$

*Then*

$$\mathcal{H}_\infty = \bigcap_{m \geq 1} \bigcup_{|w|=m} Q_w, \tag{6.2}$$

*$\mathcal{H}_\infty$ is compact, its connected components are precisely the end fibers $E_\omega$, and $\widehat{\mathbb{C}} \setminus \mathcal{H}_\infty$ is connected. The component quotient restricts to a continuous closed surjection $\mathcal{H}_\infty \to \mathtt{A}^{\mathbb{N}} = \{0,1\}^{\mathbb{N}}$, so the component space of $\mathcal{H}_\infty$ is the Cantor space, and*

$$g\colon \mathcal{H}_\infty \longrightarrow \mathcal{H}_\infty$$

*is a trivial two-sheeted covering of $\mathcal{H}_\infty$ over itself, semiconjugate under the quotient to the one-sided binary shift, realized by the two univalent branches $\phi_0, \phi_1$ of $f^{-q}$.*

*Proof.* For (6.2): every $E_\omega$ lies in $Q_{\omega|m}$ for all $m$, giving one inclusion. Conversely, a point $x$ of the right-hand side lies, for every $m$, in a unique depth-$m$ cylinder $Q_{w_m}$, and uniqueness forces $w_{m+1}|m = w_m$ as in Theorem 6.1; hence $x \in E_\omega$ for $\omega = \lim w_m$. No finite fiber meets the right-hand side: for $m > |u|$, $K_u$ misses every depth-$m$ cylinder (Lemma 5.1(c) and (b)). Thus (6.2) holds, and $\mathcal{H}_\infty$ is an intersection of compacta, hence compact.

The levels $\bigcup_{|w|=m} Q_w$ are finite unions of pairwise disjoint full continua and decrease in $m$, so Lemma 3.2 makes $\widehat{\mathbb{C}} \setminus \mathcal{H}_\infty$ connected. Components: the sets $\mathcal{H}_\infty \cap Q_u$ are clopen in $\mathcal{H}_\infty$ (they are the restrictions of the clopen sets $\mathcal{H}_\partial[u]$ of Theorem 5.3), and the argument of Theorem 5.3(iii) applies verbatim with only end addresses present. The quotient statement restricts Theorem 5.4; $\mathtt{A}^{\mathbb{N}} \subset \Omega_\partial$ is closed, and the restricted map is again a continuous closed surjection. Finally $\phi_a(\mathcal{H}_\infty) = \mathcal{H}_\infty \cap Q_a$ and $g(E_{a\omega}) = E_\omega$ give the two-sheeted covering of $\mathcal{H}_\infty$ over itself and the semiconjugacy to the full one-sided shift, exactly as in Theorem 5.5; here no root is removed, since $\sigma_{\mathrm{pre}}$ is total on ends. □

*Remark* 6.3. $\mathcal{H}_\infty$ is the statement most directly comparable to a "Cantor family of continua": a compact nonseparating union of full continua whose component space is $\{0,1\}^{\mathbb{N}}$ and on which the binary shift is realized by two actual univalent inverse branches. It does not identify the $E_\omega$ as small Julia sets (Non-claims 2.3(N2)); it realizes exactly the topological and dynamical skeleton.

6.3. **The gap-cell atlas.** Recall $G_0$, the component of $\operatorname{int} P \setminus (Q_0 \cup Q_1)$ containing $K_\epsilon$ (Standing Hypotheses 2.2(H4)). By Lemma 3.4 choose once and for all a closed Jordan disk $D$ with

$$K_\epsilon \subset \operatorname{int} D \subset D \Subset G_0, \tag{6.3}$$

and put $D_w = \phi_w(D)$.

**Theorem 6.4** (Gap-cell atlas)**.** *The closed Jordan cells $D_w$ are pairwise disjoint—including for ancestor–descendant pairs—and satisfy*

$$K_w \subset \operatorname{int} D_w, \qquad D_w \Subset \operatorname{int} Q_w \setminus (Q_{w0} \cup Q_{w1}), \tag{6.4}$$

$$D_w \cap \mathcal{H}_\partial = K_w, \qquad g(D_{aw}) = D_w. \tag{6.5}$$

*If $S_w \Subset \operatorname{int} D_w$ is compact and connected, then*

$$\operatorname{Fill}(S_w) \Subset \operatorname{int} D_w. \tag{6.6}$$

*Proof.* By (6.3) and (H2), $\phi_w$ carries $\operatorname{int} P$ to $\operatorname{int} Q_w$, $Q_a$ to $Q_{wa}$, and $D$ to $D_w$, which proves (6.4). For disjointness: incomparable words have disjoint cylinders containing their cells; and if $v = uat$ is a strict descendant of $u$, then $D_v$ lies in the child cylinder $Q_{ua}$ while $D_u$ misses both children of $Q_u$ by (6.4).

First identity of (6.5): fibers with address strictly extending $w$ lie in the children $Q_{w0} \cup Q_{w1}$, which $D_w$ misses; fibers with address incomparable to $w$ miss $Q_w \supset D_w$; fibers $K_u$ with $u$ a strict prefix of $w$ miss $Q_{w|_{|u|+1}} \supset D_w$ by Lemma 5.1(c); end fibers extending $w$ lie in a child of $Q_w$. What remains is $K_w \subset \operatorname{int} D_w$. The second identity follows from $g \circ \phi_a = \mathrm{id}$.

For (6.6): a collar of $\partial D_w$ together with the exterior of $D_w$ is a connected subset of the unbounded component of $\mathbb{C} \setminus S_w$; every bounded complementary component of $S_w$ is therefore contained away from that collar, inside $\operatorname{int} D_w$. □

6.4. **Cellwise replacement at arbitrary addresses.** The next theorem removes the finiteness restriction of the packet replacement used in Draft I. Replacement may occur at *every* finite address simultaneously; finiteness is required only where the connector construction of Section 7 actually uses it.

**Theorem 6.5** (Cellwise packet replacement at arbitrary addresses)**.** *Let $F \subset \mathtt{A}^{<\mathbb{N}}$ be an arbitrary—possibly infinite—set of finite addresses, and for each $w \in F$ let $L_w$ be a* nonempty *compact connected set with*

$$K_w \subset L_w \Subset \operatorname{int} D_w. \tag{6.7}$$

*Put $\widetilde{K}_w = L_w$ for $w \in F$ and $\widetilde{K}_w = K_w$ otherwise, and*

$$\widetilde{\mathcal{H}} = \bigcup_{w \in \mathtt{A}^{<\mathbb{N}}} \widetilde{K}_w \;\cup\; \bigcup_{\omega \in \mathtt{A}^{\mathbb{N}}} E_\omega. \tag{6.8}$$

*Then:*

(i) *$\widetilde{\mathcal{H}}$ is compact; its connected components are exactly the sets $\widetilde{K}_w$ ($w \in \mathtt{A}^{<\mathbb{N}}$) and $E_\omega$ ($\omega \in \mathtt{A}^{\mathbb{N}}$); the realization $\alpha \mapsto \widetilde{\operatorname{fib}}(\alpha)$ is upper semicontinuous; and the component projection $\widetilde{\pi}\colon \widetilde{\mathcal{H}} \to \Omega_\partial$ is a continuous closed surjection.*

(ii) *If in addition every $L_w$ ($w \in F$) is full, then $\widehat{\mathbb{C}} \setminus \widetilde{\mathcal{H}}$ is connected, and for every Jordan domain $U$ with $P \Subset U$ the relative complement $U \setminus \widetilde{\mathcal{H}}$ is connected.*

*The containment $K_w \subset L_w$ is used in neither (i) nor (ii); it records the enlargement semantics and may be dropped. Nonemptiness must then be retained so that the component projection remains surjective at every finite address.*

*Proof.* Define the modified approximants

$$\widetilde{\mathcal{H}}^{(m)} = \bigcup_{|u|<m} \widetilde{K}_u \;\cup\; \bigcup_{|w|=m} Q_w. \tag{6.9}$$

At each fixed $m$ this is a *finite* union—fewer than $2^m$ packets and exactly $2^m$ cylinders—regardless of the cardinality of $F$. The terms are pairwise disjoint: distinct packets lie in the pairwise disjoint cells $D_u$ (both $K_u$ and $L_u$ lie in $\operatorname{int} D_u$ by (6.4) and (6.7)); a packet $\widetilde{K}_u$ with $|u| < m$ and a cylinder $Q_w$ with $|w| = m$ are disjoint because either $u$ is a strict prefix of $w$, in which case $Q_w$ lies in a child of $Q_u$, which $D_u \supset \widetilde{K}_u$ misses by (6.4), or $u$ and $w$ are incomparable, in which case already $Q_u \cap Q_w = \varnothing$. The sets (6.9) decrease in $m$: each $\widetilde{K}_u$ with $|u| = m$ lies in $\operatorname{int} D_u \subset Q_u$, and each depth-$(m{+}1)$ cylinder lies in its parent.

The intersection of the modified approximants is exactly $\widetilde{\mathcal{H}}$: one inclusion is clear as before, and if $x \in \bigcap_m \widetilde{\mathcal{H}}^{(m)}$ lies in no packet $\widetilde{K}_u$, then at every depth it lies in a unique cylinder, the words nest, and $x$ lies in the corresponding end fiber. Compactness of $\widetilde{\mathcal{H}}$ follows, each approximant being compact.

For the component structure, upper semicontinuity, and the quotient in (i), the arguments of Theorems 5.3 and 5.4 apply verbatim once their two inputs are re-verified for the modified fibers. First, the separations: distinct packets lie in disjoint cells; a packet $\widetilde{K}_u$ misses every cylinder $Q_v$ with $v$ strictly extending $u$ (it lies in $D_u$, which misses the children of $Q_u$) and every cylinder incomparable to $u$; hence distinct modified fibers are pairwise disjoint, and the clopen sets $\widetilde{\mathcal{H}} \cap Q_u$ again consist exactly of the modified fibers with address extending $u$.

Second, the label-limit lemma: if $\alpha_j \to \alpha$ and $x_j \in \widetilde{\mathrm{fib}}(\alpha_j) \to x$, then for finite $\alpha$ the labels are eventually constant and $\widetilde{K}_\alpha$ is closed, while for an end $\alpha$ every modified fiber with address extending $\alpha|m$ lies in $Q_{\alpha|m}$ (packets via $\widetilde{K}_v \subset D_v \subset Q_v \subset Q_{\alpha|m}$), so $x \in E_\alpha$. With these two inputs, the clopen-partition components argument, the closed-graph upper semicontinuity, and the continuity-closedness of the component projection repeat word for word.

For (ii), every term of (6.9) is now full ($L_w$ by hypothesis, $K_u$ and $Q_w$ as before), so Lemma 3.2 and its localized clause apply to the decreasing packets, with the exterior continuum $\widehat{\mathbb{C}} \setminus U$ adjoined for the relative statement. □

*Remark* 6.6. No uniform diameter estimate, no coherence between the $L_w$ at different addresses, and no dynamical compatibility is required in Theorem 6.5; correspondingly, no equivariance is asserted for the replaced package—$g(L_{aw})$ need not equal $L_w$. Equivariance returns exactly for coherent families $L_w = \phi_w(S)$, which is the content of Corollary 6.7. For nonfull replacements, part (i) survives while part (ii) genuinely fails in general (Appendix B, Test B.6).

### 6.5. **Full root-gap naturality.**

**Corollary 6.7** (Full root-gap naturality)**.** *Let $S$ be a nondegenerate full continuum with*

$$K_\epsilon \subset S \Subset G_0,$$

*and put $S_w = \phi_w(S)$. Then the full package holds for the root $S$:*

(i) *$\mathcal{H}_\partial(S) = \bigcup_w S_w \cup \bigcup_\omega E_\omega$ is compact, its components are exactly the $S_w$ and $E_\omega$, the component projection onto $\Omega_\partial$ is a continuous closed surjection, and $g\colon \mathcal{H}_\partial(S) \setminus S \to \mathcal{H}_\partial(S)$ is a trivial two-sheeted covering with $g(S_{aw}) = S_w$;*

(ii) *every $S_w$ is full, $\widehat{\mathbb{C}} \setminus \mathcal{H}_\partial(S)$ is connected, and $U \setminus \mathcal{H}_\partial(S)$ is connected for every Jordan domain $U$ with $P \Subset U$;*

(iii) *there is a gap-cell atlas for $S$: a closed Jordan disk $D_S$ with $S \subset \operatorname{int} D_S \subset D_S \Subset G_0$ whose transported cells $D_{S,w} = \phi_w(D_S)$ satisfy all conclusions of Theorem 6.4 with $S_w$ in place of $K_w$;*

(iv) *$\mathcal{H}_\partial(S) \subset K(f)$ if and only if $S \subset K(f)$; under (H3) the same holds with $J(f)$.*

*Proof.* The root $S$ is compact, connected, nondegenerate, full, and satisfies the gap relation $S \subset G_0 \subset \operatorname{int} P \setminus (Q_0 \cup Q_1)$; that is, $S$ satisfies (H4) verbatim. Part (i) is Theorem 5.8(i)–(ii) with $R = S$, together with $g \circ \phi_{aw} = \phi_w$ for the displayed equivariance. Part (ii) reruns the proof of Theorem 6.1 with root $S$: that proof uses only fullness of the root, the separations, and (H2), all available. Part (iii) reruns Theorem 6.4: Lemma 3.4 applies to the nondegenerate full continuum $S \Subset G_0$ and produces $D_S$, and the proof of Theorem 6.4 uses only (6.3)-type containment, (H2), and the separations. Part (iv) is Theorem 5.8(iii). □

## 7. The finite protected carrier

The final theorem combines the package with the connector technology of Section 4. It is the dynamical payoff: finitely many protected accesses at the root transport to finitely many selected fibers, and a connected full carrier realizes them simultaneously while meeting the compactum in exactly the selected fibers. Here finiteness is genuinely used—the connector is a finite tree—in contrast to the replacement theorem, which is not so restricted.

**Lemma 7.1** (Proper protected impressions leave a port)**.** *Let $K \Subset \operatorname{int} D$ be a nondegenerate full continuum in a closed Jordan disk $D$ and let $P_{\mathrm{prot}} \subsetneq \partial K$ be compact, possibly empty. There are $x \in \partial K \setminus P_{\mathrm{prot}}$ and a Jordan arc $\alpha$ from $x$ to $\partial D$, otherwise in $\operatorname{int} D \setminus K$. Moreover, finitely many crosscut chains of $\widehat{\mathbb{C}} \setminus K$ whose impressions have union contained in $P_{\mathrm{prot}}$ can*

*be truncated so that, writing $S$ for the connected union of $K$ with the truncated relative collar tails,*

$$\alpha \cap \mathrm{Fill}(S) = \{x\}. \tag{7.1}$$

*Proof.* If $P_{\mathrm{prot}} = \varnothing$, choose any $y \in \partial K$ and skip the distance argument below (there are then no chains to truncate, $S = K$, and only the port construction is needed). Otherwise choose $y \in \partial K \setminus P_{\mathrm{prot}}$ and put $\delta = \mathrm{dist}(y, P_{\mathrm{prot}}) > 0$. Choose $z \in \operatorname{int} D \setminus K$ with $|z - y| < \delta/4$, and let $x \in K$ minimize distance to $z$. The segment $(x, z]$ misses $K$, and $|x - y| \le 2|z - y| < \delta/2$, so $x \notin P_{\mathrm{prot}}$. The complement of the full continuum $K$ in $\widehat{\mathbb{C}}$ is an open connected, hence arcwise connected, set; join $z$ to the exterior of $D$, take a simple subarc of the concatenation starting at $x$, and stop at its first hit on $\partial D$. This is $\alpha$.

The compact arc $\alpha$ misses the compact set $P_{\mathrm{prot}}$ at positive distance. For each chain, the nested compact sets $\overline{D_n}$ have intersection equal to the impression, which lies in $P_{\mathrm{prot}}$; hence sufficiently deep relative tails lie in prescribed neighborhoods of the impressions chosen to miss $\alpha$. Truncate each chain there. Finally, extending $\alpha$ beyond $\partial D$ to $\infty$ shows that every point of $\alpha$ except $x$ lies in the unbounded component of $\mathbb{C} \setminus S$, which proves (7.1). □

**Theorem 7.2** (Finite direct-decoration protected carrier)**.** *Fix a nonempty finite set $F \subset \mathtt{A}^{<\mathbb{N}}$ of addresses and finitely many (possibly zero) crosscut chains of $\widehat{\mathbb{C}} \setminus K_\epsilon$ (the* protected root chains*) whose relative terminal collar tails are pairwise disjoint and whose total impression is a proper compact subset $P_{\mathrm{prot}} \subsetneq \partial K_\epsilon$. Let $U$ be a Jordan domain with $P \Subset U$. Then the chains can be truncated and transported by the inverse branches $\phi_w$ ($w \in F$), and there exist a free port point $x_w \in \partial K_w$ and a port arc $\alpha_w$ for each $w \in F$, such that:*

(i) *the transported packet fullifications $L_w = \mathrm{Fill}\big(K_w \cup (\text{transported relative collar tails})\big)$ satisfy $K_w \subset L_w \Subset \operatorname{int} D_w$, so they lie in the pairwise disjoint gap cells of Theorem 6.4;*

(ii) *there is a compact connected full carrier $H \Subset U$ containing every selected fiber $K_w$ ($w \in F$) with singleton attachment trace $\overline{H \setminus K_w} \cap K_w = \{x_w\}$, and meeting the compactum in exactly the selected fibers:*

$$H \cap \mathcal{H}_\partial = \bigcup_{w \in F} K_w; \tag{7.2}$$

(iii) *every transported protected chain survives as a prime-end chain of $\widehat{\mathbb{C}} \setminus H$, and the survival map is injective and preserves the cyclic order of the transported family in $\widehat{\mathbb{C}} \setminus H$;*

(iv) *no local connectivity of any fiber is assumed or concluded.*

*Proof. Step 1: root port and truncation.* Apply Lemma 7.1 with $K = K_\epsilon$, the cell $D$ of (6.3), and $P_{\mathrm{prot}}$ (the lemma covers $P_{\mathrm{prot}} = \varnothing$): this yields a free point $x \in \partial K_\epsilon \setminus P_{\mathrm{prot}}$, an arc $\alpha$ from $x$ to $\partial D$ otherwise in $\operatorname{int} D \setminus K_\epsilon$, and truncations of the chains such that, for the connected union $S$ of $K_\epsilon$ with the truncated relative collar tails, (7.1) holds. Truncating further if necessary, $S \Subset \operatorname{int} D$: the sets $\overline{D_n}$ shrink to impressions on $\partial K_\epsilon \subset \operatorname{int} D$. The truncated relative tails are pairwise disjoint: by the null-chain condition (3.4), each truncated relative tail lies in the open terminal collar $D_1$ of its own chain, and each such collar is contained in its chain's original relative terminal tail $\overline{D_1} \setminus K_\epsilon$, which are pairwise disjoint by hypothesis—so distinct chains' truncated tails lie in disjoint sets.

*Step 2: transport.* For $w \in F$ put

$$S_w = \phi_w(S), \qquad L_w = \mathrm{Fill}(S_w), \qquad x_w = \phi_w(x), \qquad \alpha_w = \phi_w(\alpha).$$

Each $\phi_w$ restricts to an orientation-preserving homeomorphism of closed Jordan disks $D \to D_w$ ((H2), Theorem 6.4). Since $S \Subset \operatorname{int} D$ is compact and connected, Theorem 6.4(6.6), applied to

the root cell, first gives

$$\mathrm{Fill}(S) \Subset \operatorname{int} D.$$

Let $\Phi_w \colon \widehat{\mathbb{C}} \to \widehat{\mathbb{C}}$ be the pointed Schoenflies extension of $\phi_w|_D \colon D \to D_w$ from Lemma 3.3. Its Fill identity now applies with the required containment and yields

$$L_w = \mathrm{Fill}(\phi_w(S)) = \Phi_w(\mathrm{Fill}(S)) = \phi_w(\mathrm{Fill}(S)) \Subset \operatorname{int} D_w.$$

Therefore

$$\alpha_w \cap L_w = \phi_w\big(\alpha \cap \mathrm{Fill}(S)\big) = \{x_w\}. \tag{7.3}$$

The global homeomorphism $\Phi_w$ carries the truncated crosscut chains of $\widehat{\mathbb{C}} \setminus K_\epsilon$ to crosscut chains of $\widehat{\mathbb{C}} \setminus K_w$, preserving selected sides, the null-chain condition, and cyclic order; on the truncated tails it agrees with $\phi_w$. The displayed compact containment proves (i), and (7.3) says that $\alpha_w$ runs from $x_w$ to $\partial D_w$, otherwise in $\operatorname{int} D_w \setminus L_w$.

*Step 3: one free component.* The data (6.7) hold for the finite family $\{L_w\}_{w\in F}$ with every $L_w$ full, so Theorem 6.5(ii) applies: $U \setminus \widetilde{\mathcal{H}}$ is connected, where $\widetilde{\mathcal{H}}$ is the replaced compactum (6.8). Set the connector inventory of Section 4 as follows: atoms $A_w := K_w$ ($w \in F$); protected collars: the transported truncated relative collar tails, which are pairwise disjoint—within one fiber by Step 1 and injectivity of $\phi_w$, across fibers because they lie in the pairwise disjoint cells $D_w$; forbidden set $Z := \mathcal{H}_\partial \setminus \bigcup_{w\in F} K_w$ (every unselected finite fiber and every end fiber), which is compact because the selected fibers are clopen in $\mathcal{H}_\partial$. Then the frozen set $Q$ of (4.3) satisfies $Q \subset \widetilde{\mathcal{H}}$: atoms and relative collar tails lie in the packets $L_w$, and $Z \subset \widetilde{\mathcal{H}}$. Hence

$$U \setminus \widetilde{\mathcal{H}} \;\subset\; \Omega_{\mathrm{free}} = U \setminus Q,$$

and the connected set $U \setminus \widetilde{\mathcal{H}}$ lies in one component $\Omega_0$ of $\Omega_{\mathrm{free}}$.

The port interiors lie in that component: by (7.3) and (6.5),

$$\alpha_w((0,1]) \;\subset\; \big(\operatorname{int} D_w \setminus L_w\big) \cup \partial D_w \;\subset\; U \setminus \widetilde{\mathcal{H}},$$

because $D_w \cap \widetilde{\mathcal{H}} = L_w$ (the cell meets the replaced compactum exactly in its own packet, by (6.5) and disjointness of the cells) and $\partial D_w$ misses $\widetilde{\mathcal{H}}$. Also $\alpha_w \cap K_w \subset \alpha_w \cap L_w = \{x_w\}$ with $x_w \in \partial K_w$ free: $x = \phi_w^{-1}(x_w)$ is not in $P_{\mathrm{prot}}$ and hence $x_w$ is not an impression of a transported chain. The arcs $\alpha_w$ are pairwise disjoint, lying in the pairwise disjoint closed cells $D_w$. Thus $\Omega_0$ has simultaneous singleton ports to every atom.

*Step 4: certificate and carrier.* Theorem 4.3 yields a protected ribbon-routing certificate for the atoms $K_w$, protected collars, and forbidden set $Z$, with connector tree $T_{\mathfrak{R}}$ satisfying (4.1), and Theorem 4.2 realizes it: a compact connected full carrier $H = H_{\mathfrak{R}} \Subset U$ containing every $K_w$ with singleton attachment traces $\{x_w\}$, missing $Z$, with every transported chain surviving as a prime-end chain of $\widehat{\mathbb{C}} \setminus H$, injectively and in the cyclic order of the transported family. For the exact trace (7.2): $H \supset \bigcup_{w\in F} K_w$ by construction, and $H \setminus \bigcup_{w\in F} K_w = T_{\mathfrak{R}} \setminus \{x_w : w \in F\}$ lies in $\Omega_{\mathrm{free}} \cup \{\alpha_w(1)\}$, which misses $Z \cup \bigcup_{w\in F} K_w = \mathcal{H}_\partial$ except at the port points $x_w \in K_w$ already counted; hence $H \cap \mathcal{H}_\partial$ is exactly the union of the selected fibers. This proves (ii) and (iii). The arguments use crosscut chains and plane topology only, never a boundary parametrization of any fiber, proving (iv). □

*Remark* 7.3 (Finite and direct-scale only). Theorem 7.2 is finite and direct-scale only. It does not construct a second satellite portrait, an end-fiber return map, a uniform incidence bound, or the future fjord, dam, and attachment data of the virtual program (Non-claims 2.3). Selecting *infinitely* many fibers, or an end fiber, as atoms is not covered by the theorem—the connector is a finite tree—although the underlying packet compactum need not be finite (Theorem 6.5).

Proposition 8.1 shows that the raw marking data grow without bound as the decoration depth increases.

## 8. Scope and open problems

### 8.1. Why the remaining finiteness restriction is honest.

**Proposition 8.1** (Raw decoration-fan growth)**.** *The quadratic separator $Y^{(n-1)q+1}$ has exactly $2^n$ vertices, and the decoration construction supplies a distinct pairwise-disjoint univalent pullback $Q^v$ for every vertex* [KL08]*. Hence any marking that retains one distinct record for every source $Q^v$ has at least $2^n$ entries and admits no universal cardinality bound as $n$ varies.*

*Proof.* The vertex count and the one-per-vertex pairwise disjoint pullbacks are the source statements in [KL08]. Distinct disjoint pullbacks require distinct marked records, so the cardinality is at least $2^n$, which is unbounded in $n$. □

Thus a virtual construction over all scales requires either a uniformly bounded proof-complete subset of the raw fan or a future-faithful access compression; neither is constructed here (Non-claims 2.3(N4)). Note the division of labor after Theorem 6.5: the packet *compactum* supports arbitrary, even infinite, cellwise modification, while the protected *connector* remains finite.

8.2. **What separates this model from the virtual package.** Dudko's virtual Julia sets are described as connected hulls of Cantor families of small Julia sets carrying part of the critical orbit [Dud25]. The present package supplies the correctly typed skeleton of such an object at one scale: address space, embedded components, end fibers, the component quotient, both parent maps, root arrow, cells, arbitrary cellwise replacement, and protected carrier. What it does not supply is exactly what remains open:

(a) *Identification of ends.* The end fibers $E_\omega$ are nested puzzle intersections. They have not been identified with small Julia sets, and no renormalization structure on them is claimed ((N2)).
(b) *Analytic end dynamics.* The covering of Theorem 5.5 moves ends by deleting the first symbol; no first-return or self-return map *on* a single end fiber is constructed ((N3)).
(c) *Portraits and markings.* No second satellite portrait is constructed, and the transported prime-end data of Theorem 7.2 concern finitely many chains at finitely many finite fibers only ((N2), (N4)).

The two open problems below record the missing constructions in the typing of this paper. They are stated for the checked sources [Dud25, KKL26]; solving them would be a genuine advance independent of this paper.

**Problem 8.2** (Virtual package realization)**.** *For each actual intermediate virtual scale, construct an actual compact address space, nonempty compact connected components realizing it, and exact polynomial branch, return, parent, sheet, critical-address, escape, localization, and root-enlargement data—without using any downstream width, compactness, rigidity, or MLC conclusion. The present paper solves the one-scale direct-decoration case; the virtual case at deeper copies would pass through a disconnected intermediate satellite object and currently assumes the virtual renormalization being constructed.*

**Problem 8.3** (Marking transfer to the ends)**.** *Transfer a second satellite portrait to actual puzzle or generalized polynomial-like bases and actual iterates on the finite cylinders and end fibers of Theorem 5.3. Prove equivariance under (5.5), analytic realization at every end, the declared critical chronology, complete access and obstacle marking, terminal-return compatibility*

*(compatibility with the separate root arrow $\tau_{\mathrm{root}}$ of Remark 5.7 and the terminal cycle of Proposition 2.1; cf. Non-claims 2.3(N1)), and naturality under every authorized root enlargement (Theorem 5.8 and Corollary 6.7).*

8.3. **Relation to classical compactifications.** The address space (5.3) is the standard end compactification of the rooted binary tree, and nested-disk models of it are classical. The reader should therefore locate the contribution precisely: it is not the topology of $\Omega_\partial$ but the statements proved *about the dynamical realization*—membership in $K(f)$ and, under (H3), in $J(f)$; the exact component structure with its continuous closed component quotient; upper semicontinuity; the exact two-branch covering realization of the partial shift; the two-parent-map typing with a separate root arrow; fullness and nonseparation with the Cantor-indexed end compactum; the equivariant cell atlas with cellwise replacement at arbitrary addresses; and the finite protected carrier—together with the naturality of the compactification package under the root enlargement that the virtual program actually performs (Theorem 5.8 and Corollary 6.7). Each of these is used as a correctness criterion (a "positive control") against which a proposed virtual package can be audited field by field, and Appendix B records the boundary cases that audit must survive.

**Machine-assistance disclosure.** Drafting, verification, and build tooling for this paper: Fae Build, Anthropic Fable 5, and OpenAI GPT-5.6. No model output is treated as mathematical authority; all mathematical claims and proofs are the author's responsibility.

## Appendix A. Concordance with the program manuscript

This paper extracts, makes self-contained, and strengthens a package proved inside a larger program manuscript (*A Marked-Width Reduction of Mandelbrot Local Connectivity*, in preparation). For the reader with access to that manuscript, the concordance is:

| **This paper** | **Program manuscript** |
|---|---|
| Theorem 5.3 | Theorem 10.37 |
| Theorem 5.4 | — (new) |
| Theorem 5.5 | — (new) |
| Theorem 5.8 | Theorem 10.38 (strengthened) |
| Lemma 3.2 | Lemma 10.39 |
| Theorem 6.1 | Theorem 10.40 |
| Corollary 6.2 | — (new) |
| Lemma 3.4 | Lemma 10.41 |
| Theorem 6.4 | Theorem 10.42 |
| Theorem 6.5 | Theorem 10.43 (finite case; arbitrary $F$ new) |
| Corollary 6.7 | — (new) |
| Lemma 7.1 | Lemma 10.44 |
| Theorem 7.2 | Theorem 10.45 (exact trace new) |
| Theorem 4.2 | Theorem 10.4 |
| Theorem 4.3 | Theorem 10.8 |
| Lemma 3.1 | Lemma 10.10 |
| Lemma 2.4 | — (new) |
| Proposition 2.1 | Proposition 10.31 |
| Proposition 8.1 | Proposition 10.32 |

The manuscript numbering refers to its Section 10 (protected regulated connectors and the direct-decoration pullback compactification).

## Appendix B. Adversarial boundary cases

Each test below names a boundary case an auditor should try, and where the present proofs meet it.

**Test B.1** (Closed pullbacks touching)**.** Can $Q_0, Q_1$ meet at a precritical boundary point? Distinctness of the open pullback components is insufficient in general—closed puzzle pullbacks may touch [KL08]. The exclusion here is exact: any common point lies on the depth-$(nq+1)$ puzzle graph, is regular for $g$ by the explicitly stated graph-avoidance input (2.5), and Lemma 2.4 then bounds it to the closure of one component. Remark 2.6 records the deeper disjoint-$Q^v$ fallback.

**Test B.2** (Overlapping prime-end impressions)**.** Two distinct protected prime ends of a non-locally-connected fiber may have overlapping impressions. No statement in this paper requires disjoint closed tails: the certificate (Definition 4.1), the survival lemma (Lemma 3.7), and Step 1 of Theorem 7.2 use only pairwise disjoint *relative* collar tails $\overline{D_n} \setminus A$, which may have closures meeting inside the atom.

**Test B.3** (Empty protected family)**.** With no protected chains, $P_{\mathrm{prot}} = \varnothing$ and Lemma 7.1 still produces the port (its proof branches on this case); Theorem 7.2 then reduces to the port and connector construction with $S = K_\epsilon$.

**Test B.4** (Ancestor and descendant both selected)**.** Take $F = \{0, 001\}$. The cells $D_0$ and $D_{001}$ are disjoint even though the addresses are comparable (Theorem 6.4, the ancestor–descendant clause), the packets stay in their cells, and the carrier avoids the descendant's end material because $Z \supset \bigcup_\omega E_\omega$.

**Test B.5** (Infinite replacement family)**.** Take $F = \{0^k 1 : k \geq 0\}$, or $F = \mathtt{A}^{<\mathbb{N}}$. Theorem 6.5 gives compactness, the exact component structure, the continuous quotient, and (for full replacements) nonseparation: each depth-$m$ approximant is finite regardless of $|F|$.

**Test B.6** (Nonfull replacement)**.** Choose a Jordan curve $C \Subset \operatorname{int} D_w$ surrounding $K_w$, an arc $\eta \subset \operatorname{int} D_w$ joining $K_w$ to $C$, and set $L_w = K_w \cup \eta \cup C$. Then $L_w$ is compact, connected, contains $K_w$, and is not full. Part (i) of Theorem 6.5 still holds—compactum, components, quotient—while $\widehat{\mathbb{C}} \setminus \widetilde{\mathcal{H}}$ is disconnected by a bounded complementary component inside $C$. This separates the topological theorem from the fullness theorem.

**Test B.7** (Root enlargement touching a child)**.** If $S \cap Q_a \neq \varnothing$, the separation Lemma 5.1(c) fails for the enlarged root, and ancestor–descendant fibers need no longer be disjoint. The first-level-gap hypothesis in (5.8) is therefore sharp for the present proofs.

**Test B.8** (Root enlargement reaching the separator boundary)**.** If $S$ meets $\partial P$ but stays in $P \setminus (Q_0 \cup Q_1)$, the topological package survives (Theorem 5.8 uses only containment in $P$ and disjointness from the children), but the cell atlas and fullness theory require $S \Subset G_0$, as in Corollary 6.7; the corollary's hypothesis is not cosmetic.

**Test B.9** (Nondegenerate end fibers)**.** No proof in this paper uses shrinking cylinder diameters or assumes an end fiber is a point. Every argument—compactness, components, quotient, covering, fullness, cells, replacement, carrier—is valid for nondegenerate continuum end fibers.

**Test B.10** (No lower semicontinuity)**.** The realization is upper semicontinuous, and $\pi_\partial$ is continuous; neither implies that every point of $E_\omega$ is a limit of points of the finite fibers $K_{\omega|m}$. Lower semicontinuity is not proved and is used nowhere.

**Test B.11** (Root arrow returning to the package)**.** Nothing proved here rules out $g(K_\epsilon)\cap\mathcal{H}_\partial \neq \varnothing$. The root arrow is separate typed data (Remark 5.7), and no permanent-escape statement enters any proof (Non-claims 2.3(N1)).

**Test B.12** (Sphere extension moving infinity)**.** An arbitrary Schoenflies extension of $\phi_w$ need not commute with Fill: a sphere homeomorphism moving $\infty$ can exchange bounded and unbounded complementary components. Step 2 of Theorem 7.2 uses the pointed extensions $\Phi_w$ from Lemma 3.3, which fixes $\infty$ and does commute with Fill.

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